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Nouveau questionnaire de la société multidisciplinaire des glandes salivaires (MSGs) évaluant la xérostomie et les pathologies salivaires non tumorales : consensus international

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Maxillo-faciale
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Co-directeurs : Pr F. Marchal, Pr J. Seebach.

**« NOUVEAU QUESTIONNAIRE DE LA SOCIÉTÉ MULTIDISPLINAIRE
DES GLANDES SALIVAIRES (MSGs) ÉVALUANT LA XÉROSTOMIE ET
LES PATHOLOGIES SALIVAIRES NON TUMORALES : CONSENSUS
INTERNATIONAL »**

Thèse présentée à la Faculté de Médecine
de l'Université de Genève
afin d'obtenir le grade de Docteur en Médecine dentaire
par

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RÉSUMÉ

INTRODUCTION

La sécheresse orale touche 8% à 13% de la population générale (1,2), avec une prévalence qui augmente considérablement parmi les personnes de plus de 65 ans (3). Il est important de différencier une xérostomie, symptôme subjectif ressenti par les patients, d'une hyposalivation qui est une réelle diminution du flux salivaire mesurée grâce à la sialométrie.

Dans certains cas, la diminution de la sécrétion salivaire s'accompagne d'une sialadénite, représentée par une inflammation des glandes salivaires liée à la présence de calculs et/ou de sténoses.

La quantification de la sécheresse orale reste controversée et les traitements efficaces limités.

Un des moyens d'évaluer une diminution de la sécrétion de salive est l'utilisation de questionnaires. Malheureusement, malgré l'existence de multiples questionnaires, les résultats sont souvent discordants et difficiles à analyser.

L'objectif de cette étude était d'établir un nouveau questionnaire de consensus permettant l'évaluation du fardeau des symptômes de xérostomie et de sialadénite, ainsi que leur impact sur la qualité de vie des patients. Ce questionnaire en outre aurait permis de comparer les divers traitements utilisés dans la xérostomie.

MATÉRIAUX ET MÉTHODES

Un nouveau questionnaire dénommé MSGS (« Multidisciplinary Salivary Gland Society ») comprenant 20 questions et deux méthodes d'évaluation a été développé afin de quantifier les symptômes salivaires en relation avec la sécheresse orale et les pathologies salivaires non tumorales. La validation de ce questionnaire a été réalisée avec 199 patients atteints de pathologies salivaires (xérostomie digestive, nasale ou liée à l'âge, post radiothérapie cervico-faciale, post traitement par iode radioactif, syndrome de Sjögren, maladie à IgG4, parotidite juvénile récurrente, calculs et sténoses) et 66 sujets sains (contrôles). La cohérence de chaque élément du questionnaire, sa capacité de distinction entre les sujets sains et les patients, le temps nécessaire à le remplir et sa comparaison avec la sialométrie ont été analysés. Le questionnaire a été proposé et analysé sous deux versions : la première comprend quatre réponses détaillées tandis que la deuxième propose une échelle de réponses de 0 à 10 pour chacune des 20 questions. Les deux versions du questionnaire ont été analysées séparément dans des sous-groupes de patients et de volontaires sains afin de déterminer si l'une d'entre elles était plus appropriée et supérieure à l'autre.

RÉSULTATS

L'analyse statistique du nouveau questionnaire a mis en évidence une bonne cohérence interne de chacun de ses éléments, révélant ainsi sa pertinence. De plus il existait une différence significative entre les scores des sujets contrôles et des patients qui présentaient des pathologies salivaires et/ou une sécheresse orale. Les deux versions du questionnaire ont montré qu'il n'y avait pas différence statistiquement significative. Le temps moyen pour compléter le questionnaire était de 5,23 minutes en moyenne pour la version comprenant des réponses détaillées et 5,65 minutes en moyenne pour la version comprenant une échelle visuelle analogique de 0 à 10.

Les coefficients de fiabilité étaient de 90% pour la version du questionnaire comprenant des réponses détaillées et de 92% pour la version comprenant l'échelle visuelle analogique de 0 à 10.

CONCLUSION

Ce nouveau questionnaire a prouvé sa validité pour l'évaluation de la sécheresse orale et des pathologies salivaires. Ce dernier a été validé par la majorité des membres de la société internationale des glandes salivaires ce qui lui vaut la dénomination de « multidisciplinary salivary gland society (MSGS) questionnaire ». L'objectif prochain est d'implémenter ce questionnaire dans de nombreux autres pays afin de l'utiliser de manière consensuelle parmi l'ensemble des cliniciens impliqués dans la prise en charge des patients qui se présentent avec une sécheresse orale et / ou des pathologies salivaires. Ceci permettra de mieux évaluer et comparer les divers traitements utilisés dans la xérostomie et la sialadénite chronique.

SUMMARY

INTRODUCTION

Oral dryness occurs affects 8% to 13% of the general population (1,2) with an increasing prevalence among people over 65 years (3). It is important to differentiate xerostomia which is a subjective symptom experienced by patients, from hyposalivation which is a real decrease in the salivary flow, measured by sialometry.

Salivary flow reduction may be accompanied by sialadenitis which is defined as an inflammation of the salivary glands associated with the presence of stones and/or strictures.

The assessment of oral dryness remains controversial and effective treatments are limited.

One method to assess decreased salivary flow is through the use of questionnaires. Unfortunately, despite the existence of numerous questionnaires, the results remain inconsistent and difficult to analyze.

The objective of this study was to establish a new consensus questionnaire to assess the burden of the xerostomia and sialadenitis symptoms as well as their impact on the quality of life of patients. This questionnaire would also allow a comparison of all the treatments used for xerostomia.

MATERIALS AND METHODS

A novel Multidisciplinary Salivary Gland Society (MSGS) questionnaire consisting of twenty questions and two scoring systems was developed in order to quantify oral dryness and sialadenitis symptoms. Validation of the questionnaire was carried out with 199 patients suffering from salivary pathologies (digestive, nasal, or age-related xerostomia, post radiation therapy, post radioiodine therapy, Sjögren's syndrome, IgG4 disease, recurrent juvenile parotitis, stones and strictures) and 66 healthy subjects (controls). The coherence of the questionnaire's items, its reliability to distinguish patients from healthy subjects, its comparison with sialometry and the time required to complete it were analyzed.

The MSGS questionnaire was assessed in two versions: the first including four detailed answers and the second including a visual analogue scale from 0 to 10 for each of the twenty items. These two versions were analyzed separately in subgroups of patients and controls to determine whether one would be more appropriate and superior to the other.

RESULTS

The statistical analysis of our new MSGS questionnaire showed good internal coherence of the items, indicating its pertinence. Also, there was a significant difference between the scores of the healthy subjects and those of the patients suffering from salivary pathologies. The two versions of our MSGS questionnaire revealed no statistically significant difference. The average time to complete the detailed answers and the visual analogue scale versions amounted, respectively, 5.23 minutes and 5.65 minutes. The scale reliability coefficients amounted 90% for the detailed answers version and 92% for the visual analogue scale version.

CONCLUSION

Our new MSGS questionnaire has proven its validity for the evaluation of oral dryness and salivary pathologies. It has been validated by the majority of the members of the International Salivary Gland Society, which allowed its name "Multidisciplinary Salivary Gland Society (MSGS) questionnaire". Our following objective is to incorporate our MSGS questionnaire in many countries in order to use it consensually among all clinicians involved in the care of patients with oral dryness and/or salivary pathologies. This will allow a better evaluation and comparison of the various treatments used in xerostomia and sialadenitis.

INTRODUCTION

La xérostomie a été décrite pour la première fois par Critchley en 1932 dans un rapport de cas (1). Toutefois, le premier questionnaire évaluant la sécheresse orale n'a été publié qu'en 1976 par Guy (2). Il est important de différencier la xérostomie qui est un symptôme purement subjectif rapporté par les patients et mesuré grâce à des questionnaires évaluant la qualité de vie, de l'hyposalivation qui représente une réduction objective du flux salivaire mesurée grâce à la sialométrie.

Pour rappel, chaque personne adulte produit un à deux litres de salive par jour sous la régulation du système nerveux autonome. Le système nerveux parasympathique agit au travers de l'acétylcholine qui se lie aux récepteurs muscariniques en stimulant la production de salive riche en mucine, tandis que le système nerveux sympathique, au travers de la noradrénaline et de l'adrénaline qui se lient aux récepteurs bêta adrénergiques, stimule la production d'une salive séreuse.

Le flux salivaire n'est pas linéaire et dépend du rythme circadien ainsi que de la stimulation des glandes salivaires (prise d'aliments, prothèses dentaires, chewing-gum par exemple). Le flux salivaire normal au repos durant la journée est d'environ 0,3ml/minute tandis qu'à la stimulation des glandes salivaires, ce dernier s'élève à 2ml/minute (3). Durant la nuit la sécrétion salivaire diminue aux alentours de 0,1ml/minute.

La salive se compose de 99% d'eau. Le pourcentage restant représente des substances organiques et inorganiques. Les substances organiques sont des protéines telles que des enzymes (lipase, amylase par exemple), des mucines, des immunoglobulines, l'urée, le cholestérol et des hormones, tandis que les substances inorganiques sont le sodium, le potassium, le calcium, le phosphate, le chlore, le magnésium, le bicarbonate et le zinc.

La salive a de nombreux rôles : un rôle prépondérant dans la digestion des aliments grâce à la présence d'enzymes et la formation d'un bolus alimentaire, la discrimination et la perception des différents goûts, l'homéostasie microbienne de par son rôle bactériostatique et antimicrobien. La salive possède aussi une action au niveau des dents : elle est saturée en calcium et en phosphate permettant la minéralisation de l'émail dentaire ; elle comporte aussi une capacité tampon permettant la prévention de l'érosion dentaire et de la formation de la pellicule dentaire ; enfin elle favorise l'élimination des aliments et des bactéries de la surface des dents évitant ainsi la formation de caries.

La salive apporte aussi une lubrification et une hydratation des muqueuses orales et favorise leur guérison en cas de lésions. (4)

La prévalence de la xérostomie dans l'ensemble de la population varie entre 8% (5) et 13% (6) selon les études. Une étude prospective australienne sur 1'222 participants a démontré une augmentation de la prévalence de la xérostomie sur une période de trois ans (5). Ceci dénote du lien étroit entre la

sècheresse orale et l'âge avancé (prévalence de 33% dans une population de plus de 68 ans (7)). Hormis l'âge avancé, la sècheresse orale comporte de nombreuses étiologies parmi lesquelles la prise de nombreux médicaments, dont certains utilisés à grande échelle, la radiothérapie cervico-faciale de plus de 50 Grays ainsi que le syndrome de Sjögren sont les plus connus. Les médicaments incriminés dans la sècheresse orale sont les traitements anticholinergiques (antidépresseurs tricycliques, antipsychotiques, antihistaminiques, anticholinergiques pour l'hyperactivité vésicale et bronchodilatateurs), les sympathomimétiques (antidépresseurs inhibiteurs des monoamines oxydases et inhibiteurs de la recapture de la sérotonine, supprimeurs d'appétit, décongestionnants, bronchodilatateurs B2 agonistes, myorelaxants), les antihypertenseurs (inhibiteurs de l'angiotensine, antagonistes du récepteur à l'angiotensine II, bloqueurs des canaux calciques, antagonistes alpha et bêta adrénergiques, agonistes alpha 2, diurétiques), les opioïdes et benzodiazépines, les agents antimigraineux ainsi que les thérapies du SIDA. La prise des médicaments mentionnés et plus particulièrement la polymédication peuvent quadrupler la prévalence de la xérostomie. (4)

Il existe aussi de nombreuses étiologies moins connues de la sècheresse orale, telles que le traitement par radio-iodine pour certaines affections de la thyroïde (notamment tumorales etc), certaines maladies systémiques, la présence de sténoses et/ou de calculs salivaires, la déshydratation, l'anxiété et le stress. Les maladies systémiques en lien avec la sècheresse orale sont nombreuses et peuvent se subdiviser en maladies inflammatoires chroniques (arthrite rhumatoïde, parotidite récurrente de l'enfant, lupus érythémateux systémique, sarcoïdose, cirrhose biliaire primaire), pathologies endocrines (diabète, hyper ou hypothyroïdie), maladies neurologiques (dépression, maladie de Parkinson), maladies génétiques (agénésie des glandes salivaires, dysplasie ectodermique, syndrome de Prader-Willi, syndrome de Down, polyneuropathie amyloïde familiale, maladie de Gaucher, syndrome de Papillon-Lefèvre, hémochromatose héréditaire), pathologies métaboliques (insuffisance rénale chronique, boulimie, anémie, abus d'alcool), maladies infectieuses (SIDA, hépatite C) ainsi que la fibromyalgie, la pancréatite chronique et la maladie du greffon contre l'hôte. (4)

Enfin, il est intéressant de souligner une prévalence plus importante de la xérostomie chez les femmes en comparaison aux hommes. (8)

La xérostomie se mesure grâce à des questionnaires donnés aux patients qui évaluent leur qualité de vie en lien avec leurs symptômes de sècheresse orale.

Lors de l'examen clinique intra-oral il est déjà possible de mettre en évidence une sècheresse orale. Challacombe et al., en 2011 (9) ont mis en place un score clinique de sècheresse (« clinical oral dryness

score CODS ») évaluant dix éléments de la muqueuse intra-orale qui, s'ils sont altérés, suggèrent une base physiologique à la xérostomie ressentie par le patient. Les caractéristiques décrites par le CODS sont : le miroir dentaire qui colle aux muqueuses buccales et à la langue, la présence d'une salive mousseuse, l'absence de salive au niveau du plancher buccal, une perte des papilles linguales, la perte de granité gingival, une apparence vitreuse des muqueuses orales et en particulier du palais, une langue plicaturée, la présence de caries cervicales sur plus de deux dents, et la présence de débris au palais (hormis sous une prothèse dentaire).

Afin d'obtenir une évaluation plus objective, la majorité des cliniciens réalisent aussi une sialométrie dans le but de déterminer si une réelle diminution du flux salivaire est présente et pour pouvoir la quantifier. Les méthodes de sialométrie sont nombreuses, tout comme leur durée et le seuil établi pour diagnostiquer une baisse du flux salivaire. De plus le moment de la journée où la sialométrie est réalisée peut induire un biais, la sécrétion salivaire étant rythmée par le cycle circadien. Des éventuelles consommations de nourriture, de boisson ou de tabac peuvent aussi stimuler ou inhiber la sécrétion salivaire, et de ce fait induire un biais.

Il existe quatre méthodes de sialométrie : il est possible de mesurer le flux salivaire au repos ainsi que le flux salivaire sous stimulation des glandes ; de plus la sialométrie peut être non sélective en mesurant de manière aspécifique l'ensemble de la salive présente en bouche, tout comme elle peut être sélective en mesurant le flux salivaire spécifique d'une paire de glandes salivaires. La sialométrie non sélective est simple

à réaliser et peu invasive, toutefois elle est moins précise qu'une collection spécifique d'une paire de glandes. La sialométrie non sélective peut se réaliser par drainage, en laissant la salive s'écouler hors de la bouche du patient dans un récipient tout en lui demandant de pencher la tête en avant et de ne pas avaler sa salive. Il est aussi possible de demander au patient de cracher dans un récipient (Figure 1) ou d'aspirer la salive présente en bouche (Figure 2). Enfin la mise en place de cotons pré-pesés aux orifices d'aboutement des glandes constitue la méthode tampon (Figure 3). Cette dernière procédure est



Figure 1 (10)

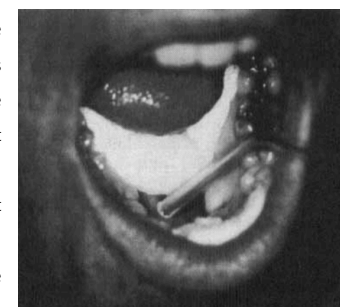


Figure 2 (11)



Figure 3 (12)

la plus répandue pour la collection de salive chez les adultes, de par sa simplicité aussi bien pour le patient que pour le clinicien. La quantification de la salive récoltée se réalise en général par la pesée du récipient / des cotons avant et après la collection salivaire.

D'autres méthodes additionnelles ont aussi été décrites dans la littérature :

En 2009, Kanehira et al (13) ont proposé l'utilisation d'un papier filtre chromatographique composé d'amidon soluble et d'iodure de potassium à un ration de 3 :1. Ce dernier est posé au centre du plancher buccal dans la région sublinguale pendant deux minutes, puis retiré. Le colorant composé de peroxyde d'hydrogène à 31% est ajouté goutte par goutte sur la surface du papier filtre et une coloration bleue apparaît, traduisant la réaction entre l'iodure de potassium et l'amidon si peu de salive était présente. Si cette dernière était présente en quantité suffisante, elle aura alors provoqué l'écoulement d'iodure de potassium hors du papier filtre et ce dernier ne réagira pas avec le colorant.

Le test de la plaquette ou « wafer test » a été décrit par Sanchez-Guerrero et al en 2002 (14). Cette plaquette ronde de 3.7cm de diamètre et un peu plus d'1mm d'épaisseur se compose de farine de blé. Elle est déposée sur le dos de la langue du patient à qui on demande alors de refermer la bouche sans mastiquer ni avaler. Le temps de dissolution complet de la plaquette est comptabilisé jusqu'à 15 minutes, au-delà desquelles le test est arrêté.

Ces deux dernières méthodes restent anecdotiques et sont très peu répandues parmi les cliniciens qui s'intéressent aux pathologies salivaires.

La saliométrie spécifique d'une paire de glandes salivaires et une procédure plus invasive nécessitant un savoir-faire de la part du clinicien et impliquant des dispositifs spécifiques.

Le dispositif de Lashley (11), de Carlson-Crittenden (15) ou de Curby (16) (Figure 5, 6) a été conçu pour prélever spécifiquement le flux salivaire de la glande parotide. Il se constitue d'un godet en plastique



Figure 4 (14)

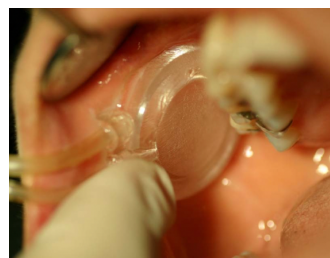


Figure 5 (16)



Figure 6 (16)

ou en métal qui se place à l'orifice du canal de Sténon. Ce dernier est connecté à la fois à une chambre interne qui permet le transport de la salive récoltée au récipient de collection ainsi qu'une chambre externe qui est attachée à un dispositif de succion (une aspiration ou une ampoule en caoutchouc) par un tube en plastique. (10)

Plusieurs dispositifs ont été publiés dans la littérature afin de prélever sélectivement la salive des glandes sous-mandibulaires et sous-linguales aux orifices des Warthon.

En 1985, Fox et al., (17) ont décrit une collection salivaire sélective aux ostiums de Warthon avec une micropipette par succion légère.

En 1997, Wolff et al., (18) ont proposé l'utilisation d'un dispositif un peu plus élaboré afin de collecter la salive aux ostiums de Warthon (Figure 7). Il se compose d'un tube collecteur de cellulose butyrate acétate stérilisable et malléable pour ajuster sa forme à l'anatomie intra-orale, relié à une chambre tampon en polycarbonate qui possède plusieurs ouvertures : la première sur le bas est directement reliée au tube de stockage de la salive, la deuxième en haut est reliée à un système d'aspiration sous forme d'une pompe à vide et la troisième est une ouverture simple afin de contrôler manuellement la succion.



Figure 7 (18)

Le test de Schirmer modifié (19), adapté du test original créé pour quantifier la sécheresse oculaire, utilise un papier filtre calibré par intervalles de 1mm jusqu'à 35mm et imprégné d'un colorant bleu à la pointe de la mèche arrondie (Figure 8). Lorsque celle-ci rentre en contact avec de l'humidité, le colorant bleu remonte le papier filtre et sa longueur peut se lire avec les intervalles désignés. Cette méthode est simple à réaliser pour le clinicien, peu invasive pour le patient et peu coûteuse.



Figure 8 (19)

La majorité des auteurs recommandent l'isolation des ostiums où les prélèvements de salive seront réalisés, par la mise en place de compresses.

Les ségrégeateurs ont été décrits dans la littérature de manière anecdotique en 1996 afin d'obtenir un prélèvement sélectif des glandes salivaires sublinguales et sous-mandibulaires (Figure 9) mais aussi des

glandes salivaires accessoires du palais (*Figure 9*). Ces dispositifs sont individuellement adaptés et se placent en bouche au niveau du plancher buccal et/ou du palais pendant 30 minutes. (20)

La méthode de collection salivaire sélective ayant le moins de biais de procédure reste la cannulation de chaque canal excréteur (aux ostiums de Sténon et Warthon), toutefois ce procédé est invasif pour le patient et nécessite un clinicien expérimenté, ce qui la rend très peu populaire.

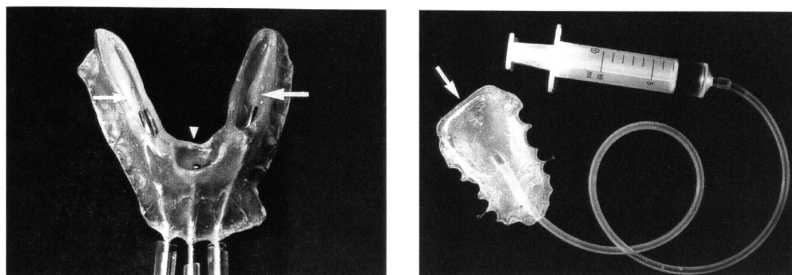


Figure 9 (20)

Certains auteurs ont décrit des méthodes de collection salivaire depuis les glandes salivaires accessoires grâce à une pipette ou un papier filtre qui sont déposés sur la surface interne des lèvres, le palais ou la muqueuse buccale puis retirés et analysés en fonction de leur humidification et ensuite convertis en flux salivaire. (21)

Le flux salivaire stimulé se réalise par stimulation gustative ou mécanique. La stimulation gustative est habituellement la plus utilisée; elle consiste en l'application de quelques gouttes d'acide citrique à 2% directement ou grâce à un papier filtre sur le dos de la langue dans le but de stimuler la sécrétion salivaire. La stimulation mécanique s'effectue par la mastication d'un chewing-gum à base de polyvinyl d'acétate ou de paraffine pendant un nombre de cycles défini, avant de récolter la salive. (11)

Le test de Saxon utilise une éponge stérile pré-pesée de 10x10cm que le patient doit mastiquer pendant deux minutes. Ensuite l'éponge est retirée de la bouche et de nouveau pesée afin d'en déduire la quantité de salive récoltée. (22)

La méthode gustative est la plus répandue dans la littérature de par sa simplicité d'exécution et l'absence de biais lié à la variation de la force masticatoire entre les patients.

Selon la revue systématique de Löfgren et al., (23), approximativement 70% des cliniciens effectuent des sialométries au repos et de manière non sélective. L'utilité d'une sialométrie spécifique d'une paire de glandes ainsi qu'une sialométrie au repos et sous stimulation a été étudiée par Vissink et al., en 2001 en relation avec le syndrome de Sjögren (24). La mesure du flux salivaire au repos et sous stimulation

permet la différenciation entre les pathologies auto-immunes où les deux valeurs de flux salivaire seront diminuées en opposition à une sécheresse orale liée à la prise de médicaments où le flux salivaire au repos sera réduit tandis que le flux salivaire stimulé restera normal. De plus la mesure du flux salivaire au repos permettrait un diagnostic précoce du syndrome de Sjögren où seul la sialométrie au repos serait diminuée tandis que la sialométrie stimulée serait encore normale. Par ailleurs, la sialométrie sélective permettrait aussi la pose d'un diagnostic précoce de syndrome de Sjögren étant donné que les glandes ne sont pas toutes atteintes au même stade de la maladie et qu'une glande encore fonctionnelle pourrait alors masquer l'hypofonction d'une autre.

Vissink et al., ont aussi mis en évidence des différences dans la composition de la salive entre les patients atteints d'un syndrome de Sjögren avant même qu'ils aient une réduction du flux salivaire et les sujets contrôles. En effet, les concentrations salivaires de sodium, chlore, albumine et IgA sont augmentées tandis que la concentration de phosphate est réduite chez les patients atteints d'un syndrome de Sjögren. Ceci évoquerait ainsi l'utilité de la sialochimie dans le diagnostic précoce du syndrome de Sjögren. (24)

Quant au seuil de flux salivaire qui permet la distinction entre une sécrétion salivaire normale ou acceptable et la présence d'une hyposalivation, il n'existe pas de consensus dans la littérature, hormis pour le syndrome de Sjögren où les critères diagnostics de la société américaine et européenne ont établi un seuil inférieur ou égal à 1,5ml/15 minutes pour le flux salivaire au repos afin de pouvoir diagnostiquer une hyposalivation (25). Ce même consensus d'expert estime que la sialométrie doit durer 15 minutes afin d'avoir des résultats plus fiables, qui ne pourraient pas être obtenus si la durée était raccourcie.

Il est intéressant de souligner qu'au sein même des Hôpitaux Universitaires de Genève, la durée de la sialométrie ainsi que le seuil de diagnostic d'hyposalivation sont différents. En effet dans le service d'Otorhinolaryngologie, une valeur de sialométrie non sélective au repos, inférieure à 2g en 6 minutes permet de poser le diagnostic d'hyposalivation, tandis que dans le service d'Immunologie et d'Allergologie une valeur de sialométrie non sélective au repos, inférieure à 0,5g pour une durée de 5 minutes est pathognomonique d'une hyposalivation.

Il n'existe pas de consensus quant au moment de la journée où la sialométrie devrait être réalisée. La majorité des cliniciens recommandent de ne pas manger, boire ou fumer une à deux heures avant la collection salivaire et de préférer la fin de matinée car le flux salivaire sera plus important qu'en début de matinée. (26)

Le bilan d'une sécheresse orale comprend de nombreux éléments comme décrits ci-dessus : un questionnaire adressé au patient relatant sa qualité de vie en lien avec ses symptômes permet d'avoir une première évaluation subjective de ses plaintes. Ensuite l'anamnèse est d'une importance primordiale afin de mettre en évidence d'éventuelles causes connues en rapport avec la symptomatologie du patient. L'examen clinique intra-oral ainsi qu'une sialométrie permettent une évaluation plus objective afin de poser le diagnostic d'hyposalivation.

Plus de 100 questionnaires évaluant la sécheresse orale sont présents dans la littérature ; certains étant spécifiquement orientés pour les patients ayant bénéficiés d'une radiothérapie cervico-faciale, ou étant diagnostiqués avec un syndrome de Sjögren par exemple, tandis que d'autres restent plus généraux. Parmi l'ensemble d'entre eux, uniquement deux se démarquent en évaluant la xérostomie et les pathologies salivaires : le « Xerostomia Inventory (XI) (27) » et le « Chronic obstructive sialadenitis score (COSS) (28) ».

Le XI a été publié en 1999 et comprends onze questions sur les symptômes salivaires et quatre questions additionnelles concernant la paresthésie buccale médicalement inexpliquée ou « burning mouth syndrome ». Cinq options de réponse sont possibles allant de « jamais » à « très souvent ». Ce questionnaire permet une bonne évaluation des symptômes de sécheresse buccale et a été validé et beaucoup utilisé dans la littérature depuis sa première publication (29, 30). Toutefois il n'est pas exhaustif quant aux symptômes de xérostomie et de sialadénite.

Le COSS « Chronic obstructive sialadenitis symptoms questionnaire » a été publié la première fois en 2016 (28), dans le but d'évaluer l'effet de la sialendoscopie sur les pathologies salivaires et la xérostomie. Il contient vingt questions concernant les symptômes du dernier mois, avec des réponses allant de 0 à 10 selon l'échelle visuelle analogique. C'est le premier questionnaire qui adresse l'ensemble des caractéristiques liées à la sialadénite, toutefois des éléments essentiels concernant la symptomatologie de la xérostomie sont manquants.

Le XI et le COSS restent ainsi incomplets pour une analyse exhaustive des symptômes de sécheresse orale et de sialadénite. Par ailleurs, aucun questionnaire existant n'est utilisé de manière consensuelle et remplit tous les critères en lien avec l'ensemble des étiologies possibles d'une sécheresse orale et de pathologies salivaires non tumorales (calculs et sténoses d'origine idiopathiques). C'est pour cette raison que nous avons développé un questionnaire de consensus qui permettrait une évaluation exhaustive de ces patients ayant des plaintes salivaires.

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ARTICLE SOUS FORMAT ORIGINAL

Novel Multidisciplinary Salivary Gland Society (MSGS) Questionnaire: An International Consensus

Multidisciplinary Salivary Gland Society Questionnaire

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Institution where the work was done:

- Concepts have been discussed with various authors from different countries.
- Patient study was conducted in Geneva:
 - o University Hospital of Geneva
 - Department of Maxillofacial Surgery and Oral Medicine and Pathology
 - Division of Clinical Immunology and Allergy
 - Department of Otorhinolaryngology-Head and Neck Surgery
 - o Pr Marchal's private practice
- The questionnaire modifications / improvements have been worked on during the International Sialendoscopy Conference in Dubai (January 2020)
- Text has been amended and corrected by all authors from their respective countries via email exchanges

Abstract

Objectives: First, the establishment and validation of a novel questionnaire documenting the burden of xerostomia and sialadenitis symptoms, including quality of life. Secondary, to compare two versions regarding the answering scale (proposed developed answers Q3 vs 0-10 visual analogue scale Q10) of our newly developed questionnaire, in order to evaluate their comprehension by patients and their reproducibility in time.

Materials and methods: A MSGS questionnaire consisting of 20 questions and 2 scoring systems was developed to quantify symptoms of dry mouth and sialadenitis. Validation of the MSGS questionnaire was performed on 199 patients with salivary pathologies (digestive, nasal or age related xerostomia, post-radiation therapy, post radioiodine therapy, Sjögren's syndrome, IgG4 disease, recurrent juvenile parotitis, stones and strictures) and a control group of 66 healthy volunteers. The coherence of the questionnaire's items, its reliability to distinguish patients from healthy volunteers, its comparison with unstimulated sialometry and the time to fill both versions were assessed.

Results: The novel MSGS questionnaire expressed good internal coherence of the items indicating its pertinence: the scale reliability coefficients amounted to a Cronbach's alpha of 0.92 for Q10 and 0.90 for Q3. The time to complete Q3 and Q10 amounted respectively to 5.23' (+/- 2.3') and 5.65' (+/- 2.64') for patients and to 3.94' (+/- 3.94') and 3.75' (+/- 2.11') for healthy volunteers. The difference between the Q3 and the Q10 was not significant.

Conclusion: We present a novel self-administered questionnaire quantifying xerostomia and non-tumoral salivary gland pathologies. We recommend the use of the Q10 version as its scale type is well known in the literature and its translation for international use will be more accurate.

Introduction

Salivary gland complaints are known to have a major effect on the quality of life of patients. In almost all cases, they comprise xerostomia and / or sialadenitis.

Xerostomia is defined as subjective oral dryness in contrast to hyposalivation, which is an objective reduction of the salivary flow [1-3]. The prevalence of xerostomia in the general population varies from 8% [4] to 13 % [5]. The range of underlying medical conditions leading to hyposalivation is increasing. Whereas Sjögren's syndrome, xerostomia after radiation therapy and xerogenic medication have been known for a long time, there is a rising incidence of other pathologies leading to xerostomia including post radio-iodine treatment for thyroid cancer [6], juvenile recurrent parotitis [7], IgG4-related disease [8], and chronic sialadenitis caused by salivary stones or salivary strictures of unknown etiology [9,10].

Sialadenitis is defined as an infection of the salivary glands, which can be caused by strictures and /or calculus. Strictures defined as a shrinkage of the salivary excretory ducts are either idiopathic, or induced by bacterial and / or auto-immune diseases, and are most frequently located in the parotid glands [11,12]. Salivary stones are calcified concretions of unknown etiology, most frequently located in the submandibular gland.

Xerostomia and sialadenitis are not rare conditions and can be linked together particularly in case of chronic sialadenitis [13], especially if they are linked to auto-immune disorders [14]. Symptoms experienced by patients presenting salivary gland pathologies are assessed by questionnaires which evaluate the patient's complaints and their repercussions on the quality of life. As of today, over 100 questionnaires regarding xerostomia and oral pathologies are available in the literature, however none has been validated to assess completely both sialadenitis and xerostomia regarding any type of treatment (medical, sialendoscopic or surgical).

The aim of this paper is to propose and validate a new self-administered questionnaire, which could be used to assess salivary gland pathologies and to quantify the burden of symptoms caused by xerostomia and sialadenitis. The secondary aim of this study is to compare two versions regarding the answering scale (proposed developed answers vs 0-10 visual analogue scale) of our newly developed questionnaire, in order to evaluate their comprehension by patients and their reproducibility in time. The purpose of this new questionnaire is to evaluate the patient's global salivary glands symptoms and his quality of life before and after medical, sialendoscopic-based or surgical treatments. It does not intend to establish a standard or a threshold above which we consider the patients "pathological" or "not pathological", since by definition it is subjective and will vary among patients. Also, it has no purpose into establishing a diagnosis of the salivary gland disease.

The main hypothesis of this study would be to be able to distinguish healthy patients from patients with salivary gland disorders with our new questionnaire. The secondary hypothesis was that a questionnaire with proposed detailed answers would be better understood and have less bias regarding its answers than a 0 to 10 visual analogue scale which would be more related to the emotional variable status of patients than on their actual symptoms.

This questionnaire was presented during the 2nd International Sialendoscopy Conference in Dubai in January, 2020 with 125 faculties from 47 countries present. It was discussed by the participants, members of the International Sialendoscopy Society and Multidisciplinary Salivary Gland Society, and accepted. It was decided during the meeting to name the questionnaire after the Multidisciplinary Salivary Gland Society (MSGS). A prospective multicentric study with 27 countries is underway, as the questionnaire will be translated into different languages and implemented in many countries.

Methods

Approval for this study was obtained from the local and national Swiss ethical committee (number 2019-00253).

Novel MSGS questionnaire:

Based on the evaluation of existing oral pathologies questionnaires and our clinical experience, we developed the MSGS questionnaire. We wanted it to be as complete as possible, as short as possible, mostly easy to understand and to fill by patients.

The questionnaire is preceded by open questions regarding: tobacco consumption, current medications having effect on xerostomia (diuretics, antidepressants, B-blockers, sleeping pills, and others), diagnosis of systemic diseases or salivary gland specific diseases (Sjögren's syndrome, IgG4-related disease, juvenile recurrent parotitis/JRP), and prior radioiodine or radiotherapy treatment. The type of gland affected is also monitored.

The MSGS questionnaire contains twenty questions divided in two groups: thirteen regarding xerostomia and seven regarding sialadenitis. Questions regarding xerostomia are following: Intensity and frequency of dry mouth; saliva quality and taste; Need to moisturize during the day and the night; Ability to talk and eat, being the most disabling consequences of dry mouth in the daily life; Dryness of lips, nose and eyes, essential to assess systemic immunological disorders, anti-cholinergic drug effects or adverse effects of radioiodine treatment; and finally QOL in relation to dry mouth symptoms. Questions regarding sialadenitis are following: Feeling of tenderness; swelling during meals - typical for acute sialadenitis - or between meals being rather signs of chronic sialadenitis related swellings may occur anytime; Persistence of swellings and need to take antibiotics - indicating the severity of sialadenitis and helping the comparison before and after treatment; Pain and discomfort induced by these swellings aiming to assess the disability and QOL encountered by the patients.

We assessed two versions of the MSGS questionnaire using the same questions but different answering scales; one with four detailed possible answers (Q3, **Table 1**), the other using a visual analogue scale (Q10, **Table 2**). We hypothesized that fixed answers might add more objectivity than a zero to ten visual analogue scale, which might be influenced by pain or the mood of the patients. The Q3 version had a total score which could vary between 0 and 60 points as every question had four possible answers which were quoted from 0 to 3 points. The Q10 version had a total score which could vary from 0 to 200 points as there could be 0 to 10 points attributed to every question.

Validation method of the MSGS questionnaire:

Patients and volunteers were included in this study from the 8th of August 2017 until the 1st of October 2020. 66 healthy volunteers from acquaintances of the main authors, nurses and physicians or colleagues from the Geneva University Hospital without salivary symptoms or known salivary diseases were selected through an anamnesis including past or actual salivary gland symptoms such as pain, swelling in the area of the salivary glands and dry mouth. They were also asked about known medical conditions which could be linked to salivary gland disorders and the use of any medication. If any of those criteria were present, they were excluded. Regarding the healthy volunteers, the inclusion criteria were the following: absence of salivary symptoms or known salivary diseases. 199 patients visiting the salivary gland center in Geneva were recruited presenting the following conditions: digestive, nasal or age related xerostomia, post radiation therapy and radioiodine therapy, Sjögren's syndrome, IgG4 disease, juvenile recurrent parotitis, stones and strictures of unknown etiology. As their pathophysiological expression are strictures, Sjögren's syndrome, IgG4 disease, radioiodine therapy and juvenile recurrent parotitis patients were included in the stenosis group.

Participants were asked to **complete both versions Q3 and Q10 of the MSGS questionnaire**.

Time to complete both Q3 and Q10 was assessed in minutes. The aim was to highlight an eventual discrepancy regarding the time needed to complete both versions.

Stenotic pathologies (idiopathic stenosis, Sjögren's syndrome, IgG4 disease and JRP), generally linked with chronic sialadenitis, were analyzed separately using the subscore of items 1-13 regarding xerostomia. As stones refers more frequently to acute sialadenitis, these symptoms were analyzed separately using the second part of the questionnaire, being the subscore of items 14-20.

A subgroup of patients randomly assigned was asked to fill out the two versions of the questionnaire Q3 and Q10 within fifteen days after their outpatient clinic visit and asked to send it back for analysis, in order to assess the reproducibility of answering pattern in time.

Sialometry was also performed in order to test whether the scores obtained by the novel MSGS questionnaire would predict hyposalivation. The method used for sialometry was following: five by five centimeters gauze compresses were disposed over each Stensen's papillas and over both Wharton's papillas during 6 minutes and/or during 15 minutes. The compresses were weighed before and after saliva collection. We chose to assess the sialometry during 6 minutes as it is a standardized time duration used in the Geneva University Hospital, and also to extend the sialometry up to 15 minutes, as the European-American consensus regarding Sjögren's syndrome established a duration of 15 minutes for sialometry to be more accurate [15]. Although the draining and the spitting methods are the most frequent sialometry methods

used in the literature as they have been acknowledged to be reproducible and reliable [16], we chose to use the swab method in our study, as we believe it to be very difficult to ask patient with xerostomia to spit or to drain saliva in a dry mouth.

There is no sialometry consensus cutoff to differentiate normal from low salivary flow, however most authors agree that under 0.1ml/min a diagnosis of hyposalivation can be established and that values below 0.25ml/min are considered low and below normal [17-19]. Regarding these considerations repeatedly published in the literature, we defined a binary threshold for sialometry < 1.5 gram at 15' [20] and < 0.72 gram at 6' [21]. The data collection was performed using case report forms (CRFs) (Appendix 1) which contained both the Q3 and Q10 MSGS questionnaire preceded by open questions regarding the general health condition which could be linked to salivary gland disorders. The time needed to complete the questionnaire and the sialometry values were included in those CRFs. The data abstraction into an Excel file has been performed by 2 authors and then verified by the statistician.

Statistical analysis section

All the statistical analysis were done with Stata v16.0. The coherence of the questionnaire items, the time to complete both Q3 and Q10, the reliability of the questionnaire to distinguish pathological patients from healthy volunteers and its comparison with sialometry were assessed. The two variants of questionnaires Q3 and Q10 were separately analyzed for a subgroup of patients and controls. Groups were compared using one-way ANOVA and Kruskal-Wallis test with Bonferroni adjustments for multiple comparisons. Results are presented as mean $\pm$ standard deviation.

To assess the validity of the question selection process a factor analysis (FA) was performed separately for the Q10 and the Q3 questionnaires.

Analysis of the reliability of repeated questionnaires was performed using paired t-test and the coefficient of variation (100x standard deviation / mean of two repeated measures) as well as the intra-class correlation coefficient (ICC) computed, along with its 95% confidence interval (95% CI) using one-way random-effect models ("icc" Stata's command). Test-retest reliability was also tested with correlation coefficient and Pitman's test of difference in variance. We computed the scales and subscales reliability coefficient using Cronbach's alpha.

The association between our questionnaire and sialometry was analyzed using linear and quadratic regression models to predict the quantity of saliva (6' and 15') with the total score of Q10 and Q3 as the independent variables. The coefficient of determination (R^2) provides the percentage of variance explained by the models. Logistic regression was used to predict binary (yes/no) outcomes (pathological sialometry; the presence of stone/stenosis) from the Q10 and Q3 or their subscores, the results are expressed as Odds ratio (OR), which can be interpreted as risk ratio.

Results

The patient group consisted of 65.4% women and 34.6% men with a mean age of 54.5 years.

The control group consisted 65.2% women and 34.8% men with a mean age of 39.2 years. The baseline characteristics and distribution of different salivary gland pathologies is shown in

Table 3: 37.26% of the 265 participants had strictures (65.6% of unknown aetiology, 28.3% Sjögren's syndrome, 2% IgG4 disease, 2% radioiodine therapy and 2% recurrent juvenile parotitis), 14.83% had stones, 16% had undergone head and neck radiotherapy treatment, 3.42% had nasal, digestive or age-related xerostomia, 3.42% had medication xerostomia, and 25.1% were healthy controls. Stenosis includes idiopathic etiologies and disease related etiologies (Sjögren's Syndrome, IgG4-related disease, juvenile recurrent parotitis, radio-iodine treatment). As there was a reduced number of disease related stenosis, these patients were included in the stenosis group. The mean age of the participants was 50.5 years $\pm$ 16.1 with 68.3% females

149 patients and 25 healthy volunteers completed both Q3 and Q10 versions of the MSGS.

The consistency of the scores measured by the intraclass correlation coefficient (ICC), were similar for both Q3 and Q10 and their subscore. Test-retest reliability was also assessed with correlation coefficient showing similar results as with the ICC and with Pitman's tests of difference in variance that also show the equivalence of the two repeats.

The scale reliability coefficients amounted to a Cronbach's alpha of 0.92 for Q10 and 0.90 for Q3, as to the subscore for the first thirteen items, they amounted to 94.6% for Q10 and 92.6% for Q3; the subscore of the seven last items amounted to 87% for Q10 and 83.2% for Q3. This shows that patients remained consistent with their answers, and that no questionnaire was more reliable than the other. Furthermore, we can observe that the thirteen first items of both

Q10 and Q3 have a better internal coherence than the last seven items and the complete questionnaires.

Factor analysis (FA) was performed separately for the Q10 and the Q3 questionnaires (added in the statistical analysis section page 13). The clustering of the question corresponds well to our clinical grouping. Loading plot for the Q10 version revealed that the first two factors account for 89.2% of the variance. Loading plot for the Q3 version assessed that the first two factors account for 88.6% of the variance. As the questions 1, 2, 3 and 5 are those who contribute the most to the factor 1, we could assume that it concerns mostly xerostomia symptoms. Regarding the factor 2, the last seven items are the ones which are the most contributive meaning that it concerns mostly sialadenitis symptoms.

Regarding the **time to complete the MSGS questionnaire**, 99 patients and 26 healthy participants were recorded while completing the Q3, and they took respectively 5.23' ($\pm$ 2.3') and 3.94' ($\pm$ 3.94') vs 108 patients and 26 healthy volunteers who were recorded completing the Q10 and took respectively 5.65' ($\pm$ 2.64') and 3.75' ($\pm$ 2.11'). Although not significant ($p = 0.193$), the Q3 was completed slightly faster than the Q10. The time to complete the Q10 ($p = 0.0008$) and the Q3 was significantly ($p = 0.0134$) shorter for healthy volunteers in comparison to patients.

Among the 38 **participants who completed both questionnaires twice within 15 days**, the coefficient of variation was 44.7% for Q10 and 41.5% for Q3, respectively, supporting the limitations of a subjective evaluation as it is prone to vary inter-individually.

The intra-class correlation coefficients were not statistically significantly different between Q10 and Q3 regarding total scores, and the sub-scores for questions 1-13 and 14 to 20; as their 95% CI overlaps. These ranged from 85% to 95% for total and questions 1-13 and were equal

to 65% for the subscore questions 14-20. We computed the individual difference between both repeats of the same questionnaire. To compare Q10 and Q3, we divided the difference by the maximum score obtainable by each questionnaire (Q3 = 60; Q10 = 200), expressed in per cent. The average difference was 0.7% for Q10 and 1.0% for Q3, and was not statistically different ($p = 0.3022$).

Regarding **sialometry**, 92 patients and 18 healthy volunteers underwent a 6 minutes sialometry and had a mean salivary flow of respectively 3.32 g/min (+/- 1.79 g/min) and 3.95 g/l (+/- 2.56 g/min). The difference of the sialometry at 6' between patients and healthy volunteers was not statistically significant ($p = 0.2035$). 157 patients and 24 healthy volunteers underwent a 15 minutes sialometry and had a mean salivary flow of respectively 4.2 g/min (+/- 2.33 g/min) and 5.7 g/min (+/- 3.33 g/min). The difference of the sialometry at 15' between patients and healthy volunteers was statistically significant ($p = 0.0067$) and revealed higher sialometries values for healthy volunteers in comparison to patients.

The healthy participants had significantly lower Q10 and Q3 scores and higher sialometry values at 15' than the pathological groups. The difference of sialometry values was only significant between healthy volunteers and patients at 15', supposing that healthy participants need time to produce enough saliva to differentiate them from patients. Moreover the number of healthy volunteers was limited which implies to consider those results with caution.

Healthy volunteers had a mean Q10 score of 19.6 (+/- 23.4) and a mean Q3 score of 5.4 (+/- 6.4) in contrast to patients who had a mean Q10 score of 52.6 (+/- 33.8) and a mean Q3 score of 14.9 (+/- 9.3). Patients had higher scoring values in contrast to healthy volunteers, and the difference was statistically significant for Q10 ($p = 0.049$) and Q3 ($p = 0.046$).

To further validate the questionnaire, patients with stenotic and calculous sialadenitis were analyzed using the first 13 and the last 7 questions separately. Regarding the group of patients

with stenosis, assessment with the Q10 ($n = 307$) 1-13 items subscore revealed no association ($R^2 = 1.5\%$, $p = 0.110$). However, a better association was found with Q3 ($n = 290$), although not sufficient to predict a stenosis ($R^2 = 2.4\%$, $p = 0.0018$). Regarding the group of patients with calculous sialadenitis, assessment with the Q10 14-20 items subscore revealed a low positive association for calculus ($n = 254$, $R^2 = 1.5\%$, $p = 0.036$, odds ratio = 1.019, 95% CI = 1.001 to 1.04), which was slightly better with the Q3 ($n = 263$, $R^2 = 2.4\%$, $p = 0.007$, odds ratio = 1.075, 95% CI = 1.02 to 1.12). An odds ratio of 1.075 means that for a 1-point increase in the Q3 questionnaire, the risk of stenosis is increased by 7.5%. The positive association of the Q3 has a higher correlation with the presence of a calculus and a stenosis compared to Q10. The scores did not predict the presence of stenosis or calculus, thus supporting the notion that the value of the questionnaire is to quantify the degree of salivary symptoms, and not to predict a particular disease.

Regarding **the correlation between the MSGS questionnaire scores and the sialometry results**, as shown in **Figure 1**, the amount of saliva produced at 6 minutes was significantly associated with Q10 ($N = 87$; $p = 0.0038$; adjusted $R^2 = 8.4\%$) and with Q3 using a linear model ($p = 0.0013$, adjusted $R^2 = 11.5\%$). At 15 minutes, Q10 was also significantly associated with the linear model explaining 10.3% of the variance ($N = 139$; $p = 0.0001$; adjusted $R^2 = 10.3\%$). At 15 minutes, Q3 with the linear model explained 13.6% ($p = 0.0001$; adjusted $R^2 = 13.6\%$). But the coefficient of determinations R^2 being below 80% preclude any individual prediction of the sialometry results. From the logistic regression, none of the questionnaires (total score or subscore 1-13) were associated with pathological sialometry at 6 and 15 minutes (odds ratio not statistically different from 1.0; p value ranging from 0.051 to 0.08), supporting the importance of an objective evaluation to assess hyposalivation.

Discussion

Many subjective questionnaires assessing symptoms and quality of life regarding xerostomia and salivary gland diseases exist in the literature. Most of them assess either xerostomia, sialadenitis or quality of life. There is a need for a comprehensive and standardized questionnaire to characterize the symptoms of patients experiencing xerostomia and / or sialadenitis and to evaluate the effect of all treatments.

Regarding quality of life, three questionnaires are frequently addressed in the literature: the Health Survey (SF-36, SF-8) [22-23] which evaluates the quality of life in regard to the patient's general health, the Glasgow Benefit Inventory [24] which assesses quality of life in regard of treatment in the otolaryngology field and the Oral Health Impact Profile Questionnaire [25-26] which evaluates the quality of life in regard to the patient's oral health. These scores evaluate quality of life of patients but lack specific features regarding the symptoms of xerostomia and sialadenitis. The purpose of the MSGS questionnaire is to target salivary glands pathologies symptoms and their consequences on the general quality of life. We believe that our questionnaire needs to be exhaustive regarding salivary glands symptoms, but also as concise as possible to be able to be used in clinical practice. Therefore, we added only four specific items regarding quality of life (questions 12, 13, 19 and 20). Also, we did not feel necessary to complicate the study by asking the patients to fill another questionnaire focused only on quality of life as our aim is to evaluate the overall salivary gland burden.

Regarding salivary gland symptoms, two questionnaires address both xerostomia and non-tumoral salivary gland pathologies and have been systematically validated: the Xerostomia Inventory (XI) [27] and the Chronic Obstructive Sialadenitis Score (COSS) [28].

The Xerostomia Inventory (XI) [27] was first published in 1999 and comprises eleven questions on xerostomia and four additional items regarding the burning mouth syndrome. Although the XI comprises many aspects related to dry mouth and has been validated and widely used in the literature [29-30], it is not exhaustive regarding xerostomia symptoms – as it does not evaluate the ability to speak, potential taste disturbances and saliva quality -, it does not address the quality of life and does not evaluate all sialadenitis features.

The Chronic Obstructive Sialadenitis Symptoms (COSS) Questionnaire [28] was proposed in 2016 to assess in twenty questions, the effect of sialendoscopy on general salivary pathologies and dry mouth. It is the first questionnaire addressing sialadenitis features. Since its publication it has been updated regarding the results obtained with pre- and post-operative assessment [31-33]. The COSS is able to precisely assess sialadenitis features, especially before and after sialendoscopy, however, it remains incomplete to evaluate other pathologies leading to xerostomia and treatment effects.

To validate the MSGS questionnaire 199 patients and 66 healthy volunteers completed it. The **control group** had significantly lower MSGS questionnaire results than the patient's group, supporting its ability to distinguish pathological and healthy participants. **Sialometry** at 6' and 15' was also assessed on both groups. The results showed a significant negative association between Q3/Q10 questionnaires and sialometry results but the amount of variance explained was not sufficient to allow individual predictions of sialometry values and therefore to detect hyposalivation.

We assessed two versions of the MSGS questionnaire (**Q3 and Q10**) in order to evaluate if fixed answers would add more objectivity, allowing a better **reproducibility** of the score over time, however no differences were found between the answers of patients with Q3 and Q10

questionnaires when completed within an interval of several days. Moreover, the **consistency of both versions**, measured by the intraclass correlation coefficient (ICC), were similar for both Q3 and Q10. Both versions of the MSGS questionnaire show a good internal coherence confirming that the answers to the questions are consistent among themselves. This was even more accentuated for the first thirteen items of the questionnaires.

Time to complete both versions of the MSGS questionnaire was assessed and no significant difference was highlighted. Moreover, at our knowledge the time required to fill out questionnaires regarding xerostomia and salivary gland pathologies has not been investigated in the literature.

As both versions Q3 and Q10 remain statistically similar regarding their scale reliability, their ability to distinguish pathological groups from healthy controls and time to fill out, both would be adequate to use. However, in order to have only one questionnaire as a consensus, we recommend to use the Q10 version. First, the 1 to 10 visual analogue scale is widely used and accepted in the literature and has the reputation to be better understood by patients; Second, the Q10 version will be more suitable for an international use, as its translation in different languages might be more accurate than the Q3 version.

Our study has some limitations. First, we chose to use a less common approach to collect saliva, in contrast to the more frequent methods described by Navazesh and al, as we believed the swab method to be easier to conduct with patients with dry mouth. We believed- following clinical experience- it to be very difficult to ask patient with xerostomia to spit or to drain saliva from a dry mouth. Moreover, our study has a restricted number of participants, especially in regard of radioiodine therapy and auto-immune salivary gland pathologies such as Sjögren's syndrome, IgG4 disease and juvenile recurrent parotitis and radioiodine.

Therefore, future multicentric studies are needed. The translation of the MSGS questionnaire in other languages and the initiation of a prospective multicentric study using the MSGS questionnaire have been decided during the 2nd International Sialendoscopy Society Meeting. This attempt to use in a multicentric setting a standardized questionnaire in centers dealing with salivary gland diseases may facilitate improvements of it, and allow a greater consensus in the future. As for an immediate result, it will allow to carry out comparative clinical studies on various treatment modalities being medical, sialendoscopic or surgical interventions, comparing the pre-treatment scoring and the post-treatment scoring. Hopefully, as for a future result, a wide consensus around a unique screening and assessment tool might be meaningful for all patients suffering from salivary gland diseases.

Conclusion

We propose here a novel, reliable, comprehensive and self-administered questionnaire addressing benign salivary gland lesions such as xerostomia, sialadenitis, and the associated QOL of the patients.

These questionnaires have been statistically validated on a series of patients affected by different pathologies.

Even though our questionnaire was not able to predict the presence of stones or strictures, it could discriminate patients from healthy volunteers and demonstrates a good internal coherence of its items. Both versions Q3 and Q10 of the MSGS questionnaire were analysed and proved to be similar regarding their scale reliability, their ability to distinguish pathological groups from healthy controls and time to fill out.

We recommend the use of the Q10 version of the MSGS questionnaire as its scale is better understood by patients and clinicians. In addition, it is more suitable for translation as we aim to implement its use for international studies.

Further multicentric investigations are in progress in order to validate it and improve it with a larger cohort of patients in twenty-seven different countries.

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Table 1: MSGS salivary score Q3 (0-3 detailed version)

Measure	Items	Scoring
Dry mouth	<u>SINCE ONE MONTH:</u>	
	1. EVALUATE THE INTENSITY OF YOUR MOUTH DRYNESS	0. NO DRYNESS 1. MILD DRYNESS, BUT NO DISCOMFORT 2. MODERATE, IMPORTANT DISCOMFORT 3. IMPORTANT, HANDICAP FOR EVERYDAY LIFE
	2. EVALUATE THE FREQUENCY OF YOUR MOUTH DRYNESS DURING THE DAY	0. NEVER 1. OCCASIONALLY DURING THE DAY 2. FREQUENTLY DURING THE DAY 3. CONSTANTLY PRESENT DURING THE DAY
	3. EVALUATE THE QUALITY OF YOUR SALIVA	0. NORMAL (EVEN IF DIMINISHED) 1. THICKER OR MORE WATERY (SEROUS) THAN NORMAL BUT WITHOUT DISCOMFORT 2. THICKER OR MORE WATERY (SEROUS) THAN NORMAL BUT WITH DISCOMFORT 3. STICKY OR WATERY (SEROUS) OR NO SALIVA
	4. EVALUATE THE TASTE OF YOUR SALIVA	0. NORMAL 1. A BIT SALTY / SWEET / BITTER / ACID / BAD TASTE 2. MODERATE SALTY / SWEET / BITTER / ACID / BAD TASTE 3. VERY SALTY / SWEET / BITTER / ACID / BAD TASTE
	5. DO YOU FEEL THE NEED TO MOISTURE YOUR MOUTH DURING THE DAY (EITHER BY DRINKING WATER / CHEWING GUMS / OR BY USING MOISTURIZING SPRAYS)?	0. NO 1. YES, OCCASIONALLY (MANY TIMES PER DAY) 2. YES, FREQUENTLY (MANY TIMES PER HOUR) 3. YES, CONSTANTLY
	6. DO YOU WAKE UP AT NIGHT TO DRINK WATER?	0. NO 1. YES, RARELY (ONE TIME MAXIMUM) 2. YES, FREQUENTLY (2-3 TIMES PER NIGHT) 3. YES, ALWAYS (MORE THAN 3 TIMES PER NIGHT)
	7. DO YOU HAVE DIFFICULTIES TALKING?	0. NO DIFFICULTY 1. YES, SOME DIFFICULTIES, I HAVE TO MOISTURIZE OCCASIONALLY WHILE TALKING 2. YES, SIGNIFICANT DIFFICULTIES, I HAVE TO MOISTURIZE FREQUENTLY WHILE TALKING 3. YES, IMPORTANT DIFFICULTIES, I HAVE TO MOISTURIZE CONSTANTLY WHILE TALKING
	8. DO YOU HAVE DIFFICULTIES CHEWING AND SWALLOWING FOOD?	0. NO 1. YES, I NEED TO DRINK TO CHEW AND SWALLOW DRY FOOD 2. YES, I NEED TO DRINK TO CHEW AND SWALLOW MOIST FOOD , I AVOID EATING DRY FOOD 3. YES, I NEED TO DRINK TO CHEW AND SWALLOW MOIST FOOD , IT IS IMPOSSIBLE FOR ME TO EAT DRY FOOD
		0. NO

	9. DO YOU HAVE DRY LIPS? 10. DO YOU HAVE A DRY NOSE? 11. DO YOU HAVE DRY EYES? 12. ARE YOUR PHYSICAL ACTIVITIES DISTURBED BECAUSE OF YOUR DRY MOUTH? 13. EVALUATE YOUR QUALITY OF LIFE REGARDING TO YOUR DRY MOUTH	1. YES, OCCASIONALLY 2. YES, FREQUENTLY 3. YES, ALWAYS 0. NO 1. YES, OCCASIONALLY 2. YES, FREQUENTLY 3. YES, ALWAYS, I NEED TO LUBRICATE IT 0. NO 1. YES, OCCASIONALLY 2. YES, FREQUENTLY, I NEED TO LUBRICATE THEM 3. YES, ALWAYS, I NEED TO LUBRICATE THEM 0. NO 1. NO, BUT I NEED TO HAVE LIQUID WITH ME 2. YES, I EXERCISE LESS THAN BEFORE 3. YES, I AVOID ANY PHYSICAL ACTIVITY THAT MAKES ME UNCOMFORTABLE BECAUSE OF DRY MOUTH 0. PERFECT 1. SATISFYING 2. LESS SATISFYING 3. COMPLETELY UNSATISFYING, MY QUALITY OF LIFE IS HIGHLY REDUCED
salivary glands	<u>SINCE ONE MONTH :</u> 14. DO YOU EXPERIENCE A FEELING OF ITCHING /TIGHTNESS (TENSION) IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW)? 15. DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) DURING MEALS? 16. DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) BETWEEN MEALS? 17. EVALUATE THE PERSISTENCE OF THIS SWELLING 18. HOW MANY TIMES HAVE YOU HAD TO TAKE ANTIBIOTICS BECAUSE OF AN INFECTION OF THE SALIVARY GLANDS?	0. NO 1. ONE TO MANY TIMES PER YEAR 2. ONE TO MANY TIMES PER MONTH 3. ONE TO MANY TIMES PER WEEK 0. NO 1. ONE TO MANY TIMES PER YEAR 2. ONE TO MANY TIMES PER MONTH 3. ONE TO MANY TIMES PER WEEK 0. NO 1. ONE TO MANY TIMES PER YEAR 2. ONE TO MANY TIMES PER MONTH 3. ONE TO MANY TIMES PER WEEK 0. NO SWELLING 1. THE SWELLING HEAL VERY QUICKLY / SPONTANEOUSLY / AFTER A FEW HOURS 2. THE SWELLING HEAL AFTER A FEW DAYS 3. THE SWELLING HEAL AFTER A FEW WEEKS / MONTHS 0. NEVER

	19. EVALUATE THE PAIN CAUSED BY THE SALIVARY GLANDS SWELLING 20. EVALUATE THE DISCOMFORT CAUSED BY THE SALIVARY GLANDS SWELLING	1. ONE TIME PER YEAR 2. MANY TIMES PER YEAR 3. ONE TIME PER MONTH 0. NO PAIN / NO SWELLING 1. MILD PAIN 2. MODERATE PAIN 3. SEVERE PAIN 0. NO DISCOMFORT / NO SWELLING 1. MILD DISCOMFORT 2. MODERATE DISCOMFORT 3. SEVERE DISCOMFORT
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Table 2: MSGS salivary score Q10 (0-10 scale version)

Measure	Items
Dry mouth	SINCE ONE MONTH : 1. <u>EVALUATE THE INTENSITY OF YOUR MOUTH DRYNESS</u> NO DRYNESS 0 1 2 3 4 5 6 7 8 9 10 MAXIMAL DRYNESS 2. <u>EVALUATE THE FREQUENCY OF YOUR MOUTH DRYNESS DURING THE DAY</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 CONSTANTLY DURING THE DAY 3. <u>EVALUATE THE QUALITY OF YOUR SALIVA</u> NORMAL (EVEN IF DIMINISHED) 0 1 2 3 4 5 6 7 8 9 10 VERY THICK / STICKY / WATERY (SEROUS) / NO SALIVA 4. <u>EVALUATE THE TASTE OF YOUR SALIVA</u> NORMAL 0 1 2 3 4 5 6 7 8 9 10 VERY SALTY AND/OR SWEET AND/OR AND/OR BITTER AND/OR ACID AND/OR BAD TASTE 5. <u>AT WHICH FREQUENCY DO YOU FEEL THE NEED TO MOISTURE YOUR MOUTH DURING THE DAY (EITHER BY DRINKING WATER / CHEWING GUMS / OR BY USING MOISTURIZING SPRAYS)?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 CONSTANTLY 6. <u>HOW FREQUENTLY DO YOU WAKE UP AT NIGHT TO DRINK WATER ?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 VERY FREQUENTLY 7. <u>EVALUATE YOUR TALKING DIFFICULTY RELATED TO YOUR DRY MOUTH</u> NO DIFFICULTY 0 1 2 3 4 5 6 7 8 9 10 VERY IMPORTANT DIFFICULTY (CONSTANT NEED TO MOISTURIZE TO BE ABLE TO SPEAK) 8. <u>EVALUATE YOUR LEVEL OF DIFFICULTY TO CHEW AND SWALLOW FOOD</u> NO DIFFICULTY 0 1 2 3 4 5 6 7 8 9 10 VERY IMPORTANT DIFFICULTY (CONSTANT NEED TO DRINK WATER TO CHEW AND SWALLOW FOOD) 9. <u>EVALUATE THE DRYNESS OF YOUR LIPS</u> NO DRYNESS 0 1 2 3 4 5 6 7 8 9 10 MAXIMAL DRYNESS 10. <u>EVALUATE THE DRYNESS OF YOUR NOSE</u>

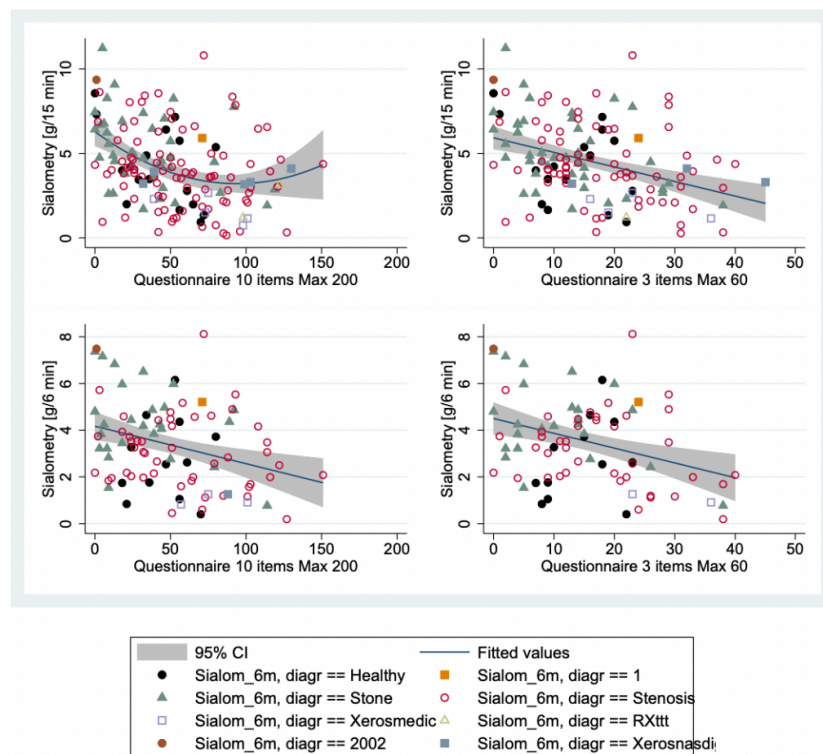
	NO DRYNESS 0 1 2 3 4 5 6 7 8 9 10 MAXIMAL DRYNESS 11. <u>EVALUATE THE DRYNESS OF YOUR EYES</u> NO DRYNESS 0 1 2 3 4 5 6 7 8 9 10 MAXIMAL DRYNESS 12. <u>ARE YOU PHYSICAL ACTIVITIES DISTURBED BECAUSE OF YOUR DRY MOUTH ?</u> NO 0 1 2 3 4 5 6 7 8 9 10 YES, I AVOID ANY ACTIVITY WHICH MAKES ME UNCOMFORTABLE BECAUSE OF MY DRY MOUTH 13. <u>EVALUATE YOUR QUALITY OF LIFE REGARDING TO YOUR DRY MOUTH</u> PERFECT 0 1 2 3 4 5 6 7 8 9 10 COMPLETELY UNSATISFYING
Salivary glands	SINCE ONE MONTH : 14. <u>AT WHAT FREQUENCY DO YOU EXPERIENCE A FEELING OF ITCHING / TIGHTNESS (TENSION) IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) ?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 EVERYDAY 15. <u>AT WHAT FREQUENCY DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) DURING MEALS ?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 AT EVERY MEAL 16. <u>AT WHAT FREQUENCY DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) BETWEEN MEALS ?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 AT EVERY MEAL 17. <u>EVALUATE THE PERSISTENCE OF THESE SWELLING</u> NO SWELLING 0 1 2 3 4 5 6 7 8 9 10 HEALING AFTER A FEW MONTHS 18. <u>HOW MANY TIMES HAVE YOU HAD TO TAKE ANTIBIOTICS BECAUSE OF AN INFECTION OF THE SALIVARY GLANDS ?</u> NEVER 0 1 2 3 4 5 6 7 8 9 10 FREQUENT INFECTIONS (> ONE TIME PER MONTH) 19. <u>EVALUATE THE PAIN CAUSED BY THE SALIVARY GLANDS SWELLING</u>

	NO PAIN	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL PAIN
20.	EVALUATE THE DISCOMFORT CAUSED BY THE SALIVARY GLANDS SWELLING												
	NO DISCOMFORT	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL DISCOMFORT

Table 3: Participant's characteristics by diagnosis compared with healthy (* p < 0.05; ** p < 0.001). N; mean ± SD.
Stenosis includes: idiopathic stenosis (n=65), Sjögren's syndrome (n=28), IgG4 disease (n=2), radioiodine therapy (n=2) and juvenile recurrent parotitis (n=2).

	Healthy	Stones	Idiopathic stenosis	Sjögren's syndrome stenosis	Other stenosis (IgG4 disease, radioiodine therapy, juvenile recurrent parotitis)	Medication related	Radiotherapy treatment	Digestive, nasal or age related	Total
	N=66	N=40	N=64	N=29	N=6	N=9	N=42	N=9	N=265
	N = 25.1%	N = 14.83%	N = 24.15%	N = 10.94%	N = 2.26%	N = 3.42%	N = 16%	N = 3.42%	N = 100%
Sex F	43 (65.2%)	23 (57.5%)	51 (79.7%)	27 (93.1%)	6 (100%)	7 (77.8%)	16 (38.1%)	6 (66.7%)	181 (68.3%)
Sex M	23 (34.8%)	17 (42.5%)	13 (20.3%)	2 (6.8%)	0 (0%)	2 (22.2%)	26 (61.9%)	3 (33.3%)	84 (31.7%)
Age (year)	66; 39.2 ± 13.3	40; 47.8 ± 13.6*	57; 55.3 ± 17.1**	19; 49.3 ± 16.4*	5; 46.1 ± 24.5	9; 69.9 ± 15.6**	42; 62.3 ± 12.0**	9; 52.0 ± 14.8	265; 50.5 ± 16.1
Total (Range 0-200)	63; 19.5 ± 23.3	34; 36.7 ± 31.3	57; 50 ± 30.7**	23; 62.3 ± 37.1**	5; 74 ± 35.6*	7; 78.4 ± 33.3**	32; 49.6 ± 33.3*	8; 76.4 ± 34.1**	229; 43.3 ± 36.6
Total (Range 0-60)	60; 5.5 ± 6.6	37; 12.2 ± 9.4*	51; 15.9 ± 8.7**	18; 18.3 ± 10.2**	6; 19.8 ± 12.6*	5; 18.4 ± 10.5*	41; 11.4 ± 7.1	4; 28.3 ± 13.6**	224; 12.3 ± 10.2
Q10 items 1 to 13 (Range 0-130)	59; 13.8 ± 15.6	32; 18.2 ± 21.1	56; 29.2 ± 25.4			7; 70.1 ± 20.3**	9; 66.0 ± 35.1**	7; 59.4 ± 31.6**	197; 30.9 ± 30.6
Q3 items 1 to 13 (Range 0-39)	59; 4.3 ± 4.7	33; 6.4 ± 6.5	49; 8.7 ± 7.2			5; 47.2 ± 29.5			
Q10 items 14 to 20 (Range 0-70)	58; 3.9 ± 11.1	31; 18.9 ± 15.9**	36; 23.5 ± 15.1**	15; 13.3 ± 7.4**	6; 12.3 ± 9	3; 17.3 ± 8.6**	6; 13.2 ± 9.2	4; 20.5 ± 11.2*	178; 7.9 ± 7.7
Q3 items 14 to 20 (Range 0-21)	55; 1.2 ± 3.3	33; 6.8 ± 5.2**	42; 7.5 ± 4.2**	15; 5.7 ± 5.5**	5; 26.8 ± 21.6*	4; 7.0 ± 14.0	3; 0.0 ± 0.0	2; 23.0 ± 12.7	161; 13.5 ± 16.1
Sialometry lasting 15min (ml/min)	16; 0.33 ± 0.21	29; 0.4 ± 0.15	50; 4.2 ± 2.3	17; 3.8 ± 2.4*	5; 2.9 ± 1.5	3; 0.07 ± 0.03*	2; 0.15 ± 0.1	6; 0.24 ± 0.03	
Sialometry lasting 6 min (ml/min)	10; 0.53 ± 0.4	20; 0.75 ± 0.3	26; 3.4 ± 1.7	10; 2.7 ± 1.2*	3; 2.4 ± 1.4	3; 0.12 ± 0.05*	No observations	2; 0.3 ± 0.12	

Figure 1: Regression curve/line of the association between our questionnaires (Q3 and Q10) and sialometry



Appendix 1: Case report form MSGS questionnaire

MSGS SALIVARY SCORE

Today's date:

Patient's name and birth date:

Please choose (if applicable) the side and the gland affected by the symptoms

- Left / Right
- Submandibular (under the lower jaw) / Parotid (in front of the ear)

I. General informations (to fill by the clinician)

- Do you **smoke** ? Yes / No If yes, how many cigarettes per day:
- Do you take the following **medications** ? Yes/No
If yes please mention them :
 - Diuretics
 - Sleeping pills
 - Anti-depressants
 - B-bloquants
 - Other
- Have you bee diagnosed with other **diseases** ?
- Have you had a **radioiodine** treatment ? Yes/No If yes when ?
What dose ?
- Have you been diagnosed with a **Sjögren's syndrome** or **IgG4 disease** ? Yes/No
If yes, when ?
- Since how long do you experience dry mouth symptoms (number of years) ?

II. Questionnaire (to fill by the patient)

Please circle the answer which characterizes you the most for each question.

A. Questions regarding dry mouth

Since one month :

1. EVALUATE THE INTENSITY OF YOUR MOUTH DRYNESS

NO DRYNESS 0 1 2 3 4 5 6 7 8 9 10 MAXIMAL DRYNESS

2. EVALUATE THE FREQUENCY OF YOUR MOUTH DRYNESS DURING THE DAY

NEVER 0 1 2 3 4 5 6 7 8 9 10 CONSTANTLY DURING THE DAY

3. EVALUATE THE QUALITY OF YOUR SALIVA

NORMAL 0 1 2 3 4 5 6 7 8 9 10 VERY THICK / STICKY /
WATERY (SEROUS) / NO SALIVA

4. EVALUATE THE TASTE OF YOUR SALIVA

NORMAL	0	1	2	3	4	5	6	7	8	9	10	VERY SALTY AND/OR SWEET AND/OR BITTER AND/OR ACID AND/OR BAD TASTE
--------	---	---	---	---	---	---	---	---	---	---	----	--

5. AT WHICH FREQUENCY DO YOU FEEL THE NEED TO MOISTURE YOUR MOUTH DURING THE DAY (EITHER BY DRINKING WATER / CHEWING GUMS / OR BY USING MOISTURIZING SPRAYS)?

NEVER	0	1	2	3	4	5	6	7	8	9	10	CONSTANTLY
-------	---	---	---	---	---	---	---	---	---	---	----	------------

6. HOW FREQUENTLY DO YOU WAKE UP AT NIGHT TO DRINK WATER ?

NEVER	0	1	2	3	4	5	6	7	8	9	10	VERY FREQUENTLY
-------	---	---	---	---	---	---	---	---	---	---	----	-----------------

7. EVALUATE YOUR TALKING DIFFICULTY RELATED TO YOUR DRY MOUTH

NO DIFFICULTY	0	1	2	3	4	5	6	7	8	9	10	VERY IMPORTANT DIFFICULTY (CONSTANT NEED TO MOISTURIZE TO BE ABLE TO SPEAK)
---------------	---	---	---	---	---	---	---	---	---	---	----	---

8. EVALUATE YOUR LEVEL OF DIFFICULTY TO CHEW AND SWALLOW FOOD

NO DIFFICULTY	0	1	2	3	4	5	6	7	8	9	10	VERY IMPORTANT DIFFICULTY (CONSTANT NEED TO DRINK WATER TO CHEW AND SWALLOW FOOD)
---------------	---	---	---	---	---	---	---	---	---	---	----	--

9. EVALUATE THE DRYNESS OF YOUR LIPS

NO DRYNESS	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL DRYNESS
------------	---	---	---	---	---	---	---	---	---	---	----	-----------------

10. EVALUATE THE DRYNESS OF YOUR NOSE

NO DRYNESS	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL DRYNESS
------------	---	---	---	---	---	---	---	---	---	---	----	-----------------

11. EVALUATE THE DRYNESS OF YOUR EYES

NO DRYNESS	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL DRYNESS
------------	---	---	---	---	---	---	---	---	---	---	----	-----------------

12. ARE YOU PHYSICAL ACTIVITIES DISTURBED BECAUSE OF YOUR DRY MOUTH ?

NO WHICH MAKES ME	0	1	2	3	4	5	6	7	8	9	10	YES, I AVOID ANY ACTIVITY UNCOMFORTABLE BECAUSE OF MY DRY MOUTH
----------------------	---	---	---	---	---	---	---	---	---	---	----	---

13. EVALUATE YOUR QUALITY OF LIFE REGARDING TO YOUR DRY MOUTH

PERFECT	0	1	2	3	4	5	6	7	8	9	10	COMPLETELY UNSATISFYING
---------	---	---	---	---	---	---	---	---	---	---	----	-------------------------

B. Questions regarding salivary glands

14. AT WHAT FREQUENCY DO YOU EXPERIENCE A FEELING OF ITCHING / TIGHTNESS (TENSION) IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) ?

NEVER	0	1	2	3	4	5	6	7	8	9	10	EVERYDAY
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15. AT WHAT FREQUENCY DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) DURING MEALS ?

NEVER	0	1	2	3	4	5	6	7	8	9	10	AT EVERY MEAL
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16. AT WHAT FREQUENCY DO YOU EXPERIENCE SWELLING IN THE AREA OF THE SALIVARY GLANDS (IN FRONT OF THE EARS OR/AND UNDER THE LOWER JAW) BETWEEN MEALS ?

NEVER	0	1	2	3	4	5	6	7	8	9	10	AT EVERY MEAL
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17. EVALUATE THE PERSISTENCE OF THESE SWELLING

NO SWELLING	0	1	2	3	4	5	6	7	8	9	10	HEALING AFTER A FEW MONTHS
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18. HOW MANY TIMES HAVE YOU HAD TO TAKE ANTIBIOTICS BECAUSE OF AN INFECTION OF THE SALIVARY GLANDS ?

NEVER INFECTIONS	0	1	2	3	4	5	6	7	8	9	10	FREQUENT (> ONE TIME PER MONTH)
---------------------	---	---	---	---	---	---	---	---	---	---	----	---

19. EVALUATE THE PAIN CAUSED BY THE SALIVARY GLANDS SWELLING

NO PAIN	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL PAIN
---------	---	---	---	---	---	---	---	---	---	---	----	--------------

20. EVALUATE THE DISCOMFORT CAUSED BY THE SALIVARY GLANDS SWELLING

NO DISCOMFORT	0	1	2	3	4	5	6	7	8	9	10	MAXIMAL DISCOMFORT
---------------	---	---	---	---	---	---	---	---	---	---	----	-----------------------

TIME NEEDED TO COMPLETE THE QUESTIONNAIRE (MINUTES) :

UNSTIMULATED SIALOMETRY VALUES:

- 6 MINUTES:
- 15 MINUTES:

ARTICLE SOUS FORMAT APRES PUBLICATION

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Novel Multidisciplinary Salivary Gland Society (MSGS)
Questionnaire: An International Consensus

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Jörg D. Seebach, MD; Francis Marchal, MD

INCLUSION CRITERIA	YES	NO
PATIENTS PRESENTING WITH ORAL DRYNESS WHICH CAN BE RELATED TO VARIOUS ETIOLOGIES (IF THE ETIOLOGY IS KNOWN, CHECK ONE OR MORE OF THE FOLLOWING ONES): - HEAD AND NECK RADIOTHERAPY - RADIOIODINE TREATMENT - SJÖGREN’S SYNDROME - IGG4 DISEASE - SALIVARY GLAND STRICTURES WITHOUT RELATION TO RADIATION - SALIVARY CALCULUS - JUVENILE RECURRENT PAROTITIS		

IS THE PATIENT ELIGIBLE FOR THE STUDY ?

YES

NO

EXIT OF THE STUDY

DATE :

REASONS :

- NON COMPLIANCE TO TREATMENT
- WILLINGNESS OF THE PATIENT
- LOST OF SIGHT
- UNDESIRABLE EVENT
- DEATH
- OTHERS :

REMARKS :

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Additional supporting information may be found in the online version of this article.

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Institution where the work was done:

- Concepts have been discussed with various authors from different countries
- Patient study was conducted in Geneva:
 - University Hospital of Geneva
 - Department of Maxillofacial Surgery and Oral Medicine and Pathology
 - Division of Clinical Immunology and Allergy
 - Department of Otorhinolaryngology-Head and Neck Surgery
 - Pr Marchal’s private practice
- The questionnaire modifications/improvements have been made during the International Sialendoscopy Conference in Dubai (January 2020).
- Text has been amended and corrected by all authors from their respective countries via email exchanges.

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Objectives: First, establishment and validation of a novel questionnaire documenting the burden of xerostomia and sialadenitis symptoms, including quality of life. Second, to compare two versions regarding the answering scale (proposed developed answers Q3 vs. 0–10 visual analogue scale [Q10]) of our newly developed questionnaire, in order to evaluate their comprehension by patients and their reproducibility in time.

Study Design: The study is a systematic review regarding the evaluation of the existing questionnaire and a cohort study regarding the validation of our new MSGS questionnaire.

Materials and Methods: A Multidisciplinary Salivary Gland Society (MSGS) questionnaire consisting of 20 questions and two scoring systems was developed to quantify symptoms of dry mouth and sialadenitis. Validation of the questionnaire was carried out on 199 patients with salivary pathologies (digestive, nasal, or age-related xerostomia, post radiation therapy, post radioiodine therapy, Sjögren's syndrome, IgG4 disease, recurrent juvenile parotitis, stones, and strictures) and a control group of 66 healthy volunteers. The coherence of the questionnaire's items, its reliability to distinguish patients from healthy volunteers, its comparison with unstimulated sialometry, and the time to fill both versions were assessed.

Results: The novel MSGS questionnaire showed good internal coherence of the items, indicating its pertinence: the scale reliability coefficients amounted to a Cronbach's alpha of 0.92 for Q10 and 0.90 for Q3. The time to complete Q3 and Q10 amounted, respectively, to 5.23 min (± 2.3 min) and 5.65 min (± 2.64 min) for patients and to 3.94 min (± 3.94 min) and 3.75 min (± 2.11 min) for healthy volunteers. The difference between Q3 and Q10 was not significant.

Conclusion: We present a novel self-administered questionnaire quantifying xerostomia and non-tumoral salivary gland pathologies. We recommend the use of the Q10 version, as its scale type is well known in the literature and its translation for international use will be more accurate.

Key Words: Chronic obstructive sialadenitis score (COSS), Multidisciplinary Salivary Gland Society (MSGS) questionnaire, oral dryness questionnaire (DMQ), sialadenitis, xerostomia.

Laryngoscope, 00:1–10, 2021

INTRODUCTION

Salivary gland complaints are known to have a major effect on the quality of life (QOL) of patients. In almost all cases, they comprise xerostomia and/or sialadenitis.

Xerostomia is defined as subjective oral dryness in contrast to hyposalivation, which is an objective reduction of the salivary flow.^{1–3} The prevalence of xerostomia in the general population varies from 8%⁴ to 13%.⁵ The range of the underlying medical conditions leading to hyposalivation is increasing. Whereas Sjögren's syndrome, xerostomia after radiation therapy, and xerogenic medication have been known for a long time, there is rising incidence of other pathologies leading to xerostomia including post radioiodine treatment for thyroid cancer,⁶ juvenile recurrent parotitis (JRP),⁷ IgG4-related disease,⁸ and chronic sialadenitis caused by salivary stones or salivary strictures of unknown etiology.^{9,10}

Sialadenitis is defined as an infection of the salivary glands, which can be caused by strictures and/or calculi. Strictures, defined as a shrinkage of the salivary excretory ducts, are either idiopathic or induced by bacterial and/or auto-immune diseases, and are most frequently located in the parotid glands.^{3,11} Salivary stones are calcified concretions of unknown etiology, most frequently located in the submandibular gland.

Xerostomia and sialadenitis are not rare conditions and can be linked together particularly in case of chronic sialadenitis,¹² especially if they are linked to auto-immune disorders.¹³ Symptoms experienced by patients presenting salivary gland pathologies are assessed by questionnaires that evaluate the patient's complaints and their repercussions on their QOL. As of today, over 100 questionnaires regarding xerostomia and oral pathologies are available in the literature; however, none has been validated to assesses completely both sialadenitis and xerostomia regarding any type of treatment (medical, sialendoscopic, or surgical).

The aim of this paper is to propose and validate a new self-administered questionnaire, which could be used to assess salivary gland pathologies and to quantify the burden of symptoms caused by xerostomia and sialadenitis. The secondary aim of this study is to compare two versions regarding the answering scale (proposed developed answers vs. 0–10 visual analogue scale [VAS]) of our newly developed questionnaire, in order to evaluate their comprehension by patients and their reproducibility in time. The purpose of this new questionnaire is to evaluate the patients' global salivary gland symptoms and their QOL before and after medical sialendoscopic-based or surgical treatments. It does not intend to establish a standard or a threshold above which we consider the patients "pathological" or "not pathological," since by definition it is subjective and will vary among patients. Also, its purpose is not establishing a diagnosis of the salivary gland disease.

The main hypothesis of this study would be to be able to distinguish healthy patients from patients with salivary gland disorders with our new questionnaire. The secondary hypothesis was that a questionnaire with proposed detailed answers would be better understood and have less bias regarding its answers than a 0 to 10 VAS, which would be more related to the emotional variable status of patients than to their actual symptoms.

This questionnaire was presented during the second International Sialendoscopy Conference in Dubai in January, 2020, with 125 faculties from 47 countries present. It was discussed by the participants, members of the International Sialendoscopy Society, and Multidisciplinary Salivary Gland Society, and accepted. It was decided during the meeting to name the questionnaire after the Multidisciplinary Salivary Gland Society (MSGS). A prospective multicentric study with 27 countries is under way, as the questionnaire will be translated into different languages and implemented in many countries.

METHODS

Approval for this study was obtained from the local and national Swiss ethical committee (number 2019–00253).

Novel MSGS Questionnaire

Based on the evaluation of existing questionnaires on oral pathologies and our clinical experience, we developed the MSGS questionnaire. We wanted it to be as complete as possible, as short as possible, and mostly easy to understand and fill by patients.

The questionnaire is preceded by open questions regarding tobacco consumption, current medications having effect on xerostomia (diuretics, antidepressants, beta-blockers, sleeping pills, and others), diagnosis of systemic diseases or salivary-gland-specific diseases (Sjögren's syndrome, IgG4-related disease, JRP), and prior radioiodine or radiotherapy treatment. The type of gland affected is also monitored.

The MSGS questionnaire contains 20 questions divided in two groups: 13 regarding xerostomia and 7 regarding sialadenitis. Questions regarding xerostomia are the following: intensity and frequency of dry mouth; saliva quality and taste; need to moisturize during day and night; ability to talk and eat (being the most disabling consequences of dry mouth in daily life); dryness of lips, nose, and eyes (essential to assess systemic immunological disorders, anti-cholinergic drug effects, or adverse effects of radioiodine treatment); and finally QOL in relation to dry-mouth symptoms. Questions regarding sialadenitis are the following: feeling of tenderness; swelling during meals (typical for acute sialadenitis) or between meals, which, being signs of chronic sialadenitis-related, may occur anytime; persistence of swellings and need to take antibiotics (indicating the severity of sialadenitis and helping the comparison before and after treatment); pain and discomfort induced by these swellings (aiming to assess the disability), and QOL encountered by the patients.

We assessed two versions of the MSGS questionnaire using the same questions but different answering scales: one with four detailed possible answers (Q3, Table I), and the other using a VAS (Q10, Table II). We hypothesized that fixed answers might add more objectivity than a 0–10 VAS, which might be influenced by pain or the mood of the patients. The Q3 version had a total score that could vary between 0 and 60 points, as every question had four possible answers which were quoted from 0 to 3 points. The Q10 version had a total score that could vary from 0 to 200 points, as there could be 0 to 10 points attributed to every question.

Validation Method of the MSGS Questionnaire

Patients and volunteers were included in this study from August 8, 2017 until October 1, 2020. Sixty-six healthy volunteers, who were acquaintances of the main authors, nurses and physicians, or colleagues from the Geneva University Hospital, without salivary symptoms or known salivary diseases were selected through an anamnesis including past or actual salivary gland symptoms such as pain, swelling in the area of the salivary glands, and dry mouth. They were also asked about known medical conditions that could be linked to salivary gland disorders and the use of any medication. If any of those criteria was met, they were excluded. Regarding the healthy volunteers, the inclusion criteria were the following: absence of salivary symptoms or known salivary diseases. One-hundred and ninety-nine patients visiting the salivary gland center in Geneva presenting the following conditions were recruited: digestive, nasal, or age-related xerostomia; post radiation therapy and radioiodine therapy; Sjögren's syndrome; IgG4 disease; JRP; and stones and

strictures of unknown etiology. As their pathophysiological expression is strictures, patients with Sjögren's syndrome, IgG4 disease, radioiodine therapy, and JRP were included in the stenosis group.

Participants Were Asked to Complete Both Versions Q3 and Q10 of the MSGS Questionnaire.

Time to complete both Q3 and Q10 was assessed in minutes. The aim was to highlight any eventual discrepancy regarding the time needed to complete both versions.

Stenotic pathologies (idiopathic stenosis, Sjögren's syndrome, IgG4 disease, and JRP) generally linked with chronic sialadenitis were analyzed separately using the subscore of items 1–13 regarding xerostomia. As stones refers more frequently to acute sialadenitis, these symptoms were analyzed separately using the second part of the questionnaire, being the subscore of items 14–20.

A subgroup of patients randomly assigned were asked to fill out the two versions of the questionnaire Q3 and Q10 within 15 days after their outpatient clinic visit and asked to send it back for analysis, in order to assess the reproducibility of answering pattern in time.

Sialometry was also performed to test whether the scores obtained by the novel MSGS questionnaire would predict hyposalivation. The method used for sialometry was the following: 5 cm $\times$ 5 cm gauze compresses were disposed over each Stensen's papillas and over both Wharton's papillas during 6 min and/or during 15 min. The compresses were weighed before and after saliva collection. We chose to assess the sialometry during 6 min, as it is a standardized time duration used in the Geneva University Hospital, and also to extend the sialometry up to 15 min, as the European–American consensus regarding Sjögren's syndrome has established a duration of 15 min for sialometry to be more accurate.¹⁴ Although the draining and the spitting methods are the most frequent sialometry methods used in the literature, as they have been acknowledged to be reproducible and reliable,¹⁵ we chose to use the swab method in our study because we believe it to be very difficult to ask patients with xerostomia to spit or to drain saliva in a dry mouth.

There is no sialometry consensus cutoff to differentiate normal from low salivary flow; however, most authors agree that below 0.1 ml/min, a diagnosis of hyposalivation can be established and that values below 0.25 ml/min are considered low and below normal.^{16–18}

Based on these considerations repeatedly published in the literature, we defined a binary threshold for sialometry of <1.5 g at 15 min¹⁹ and <0.72 g at 6 min.²⁰ The data collection was performed using case report forms (CRFs) (Appendix 1), which contained both Q3 and Q10 MSGS questionnaires, preceded by open questions regarding any general health condition that could be linked to salivary gland disorders. The time needed to complete the questionnaire and the sialometry values were included in those CRFs. The data abstraction into an Excel file was performed by two authors and then verified by the statistician.

Statistical Analysis

All statistical analyses were done with Stata v16.0. The coherence of the questionnaire items, the time to complete both Q3 and Q10, the reliability of the questionnaire to distinguish pathological patients from healthy volunteers, and its comparison with sialometry were assessed. The two variants of questionnaires Q3 and Q10 were separately analyzed for a subgroup of patients and controls. Groups were compared using one-way ANOVA and Kruskal–Wallis test with Bonferroni adjustments for multiple comparisons. Results are presented as mean $\pm$ standard deviation (SD).

TABLE I. MSGS Salivary Score Q3 (0-3 Detailed Version).		
Measure	Items	Scorings
Dry mouth	Since 1 mo:	0. No dryness
	1. Evaluate the intensity of your mouth dryness	1. Mild dryness, but no discomfort 2. Moderate, important discomfort 3. Important, handicap for everyday life 0. Never
	2. Evaluate the frequency of your mouth dryness during the day	1. Occasionally during the day 2. Frequently during the day 3. Constantly present during the day 0. Normal (even if diminished)
	3. Evaluate the quality of your saliva	1. Thicker or more watery (serous) than normal but without discomfort 2. Thicker or more watery (serous) than normal but with discomfort 3. Sticky or watery (serous) or no saliva 0. Normal
	4. Evaluate the taste of your saliva	1. A bit salty/sweet/bitter/acid/bad taste 2. Moderate salty/sweet/bitter/acid/bad taste 3. Very salty/sweet/bitter/acid/bad taste 0. No
	5. Do you feel the need to moisture your mouth during the day (either by drinking water/chewing gums/or by using moisturizing sprays)?	1. Yes, occasionally (many times per day) 2. Yes, frequently (many times per hour) 3. Yes, constantly 0. No
	6. Do you wake up at night to drink water?	1. Yes, rarely (one time maximum) 2. Yes, frequently (2-3 times per night) 3. Yes, always (more than 3 times per night) 0. No difficulty
	7. Do you have difficulties talking?	1. Yes, some difficulties, i have to moisturize occasionally while talking 2. Yes, significant difficulties, i have to moisturize frequently while talking 3. Yes, important difficulties, i have to moisturize constantly while talking 0. No
	8. Do you have difficulties chewing and swallowing food?	1. Yes, i need to drink to chew and swallow dry food 2. Yes, i need to drink to chew and swallow moist food, i avoid eating dry food 3. Yes, i need to drink to chew and swallow moist food, it is impossible for me to eat dry food 0. No
	9. Do you have dry lips?	1. Yes, occasionally 2. Yes, frequently 3. Yes, always 0. No
	10. Do you have a dry nose?	1. Yes, occasionally 2. Yes, frequently 3. Yes, always, i need to lubricate it 0. No
	11. Do you have dry eyes?	1. Yes, occasionally 2. Yes, frequently, i need to lubricate them 3. Yes, always, i need to lubricate them 0. No
	12. Are your physical activities disturbed because of your dry mouth?	1. No, but i need to have liquid with me 2. Yes, i exercise less than before 3. Yes, i avoid any physical activity that makes me uncomfortable because of my dry mouth 0. Perfect
	13. Evaluate your quality of life regarding to your dry mouth	1. Satisfying 2. Less satisfying 3. Completely unsatisfying, my quality of life is highly reduced 0. No
Salivary glands	Since 1 mo:	
	14. Do you experience a feeling of itching/tightness (tension) in the area of the	1. One to many times per year 2. One to many times per month 3. One to many times per week

(Continues)

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TABLE I. Continued		
Measure	Items	Scorings
	salivary glands (in front of the ears or/and under the lower jaw)?	
	15. Do you experience swelling in the area of the salivary glands (in front of the ears or/and under the lower jaw) during meals?	0. No 1. One to many times per year 2. One to many times per month 3. One to many times per week 0. No
	16. Do you experience swelling in the area of the salivary glands (in front of the ears or/and under the lower jaw) between meals?	1. One to many times per year 2. One to many times per month 3. One to many times per week 0. No swelling
	17. Evaluate the persistence of this swelling	1. The swelling healS very quickly/ spontaneously/after a few hours 2. The swelling heals after a few days 3. The swelling heals after a few weeks/ months 0. Never
	18. How many times have you had to take antibiotics because of an infection of the salivary glands?	1. One time per year 2. Many times per year 3. One time per month 0. No pain/no swelling
	19. Evaluate the pain caused by the salivary gland swelling	1. Mild pain 2. Moderate pain 3. Severe pain 0. No discomfort/no swelling
	20. Evaluate the discomfort caused by the salivary gland swelling	1. Mild discomfort 2. Moderate discomfort 3. Severe discomfort

To assess the validity of the question selection process, a factor analysis (FA) was performed separately for the Q10 and Q3 questionnaires.

Analysis of the reliability of repeated questionnaires was performed using paired *t*-test and the coefficient of variation (100× standard deviation/mean of two repeated measures), and the intraclass correlation coefficient (ICC) was computed along with its 95% confidence interval (95% CI) using one-way random-effect models ("icc" Stata's command). Test–retest reliability was also tested with correlation coefficient and Pitman's test of difference in variance. We computed the scale and subscale reliability coefficient using Cronbach's alpha.

The association between our questionnaire and sialometry was analyzed using linear and quadratic regression models to predict the quantity of saliva (6 and 15 min), with the total score of Q10 and Q3 as the independent variables. The coefficient of determination (R^2) provides the percentage of variance explained by the models. Logistic regression was used to predict binary (yes/no) outcomes (pathological sialometry; the presence of stone/stenosis) from Q10 and Q3 or their subscores, and the results are expressed as the odds ratio (OR), which can be interpreted as the risk ratio.

RESULTS

The patient group consisted of 65.4% women and 34.6% men, with a mean age of 54.5 years. The control group consisted 65.2% women and 34.8% men, with a mean age of 39.2 years. The baseline characteristics and distribution of different salivary gland pathologies are shown in Table III: 37.26% of the 265 participants had strictures (65.6% of unknown etiology, 28.3% Sjögren's syndrome, 2% IgG4 disease, 2% radioiodine therapy, and 2% RJP), 14.83% had stones, 16% had undergone head-

and-neck radiotherapy treatment, 3.42% had nasal, digestive, or age-related xerostomia, 3.42% had medication xerostomia, and 25.1% were healthy controls. Stenosis includes idiopathic etiologies and disease-related etiologies (Sjögren's syndrome, IgG4-related disease, JRP, radioiodine treatment). As there was a reduced number of disease-related stenosis, these patients were included in the stenosis group. The mean age of the participants was 50.5 years ± 16.1, with 68.3% females.

One-hundred and forty-nine patients and 25 healthy volunteers completed both Q3 and Q10 versions of the MSGS.

Consistency of the scores measured by the intraclass correlation coefficient (ICC), were similar for both Q3 and Q10 and their subscore. Test–retest reliability was also assessed, with the correlation coefficient showing similar results as with the ICC and with Pitman's tests of difference in variance, which also show the equivalence of the two repeats.

The scale reliability coefficients amounted to a Cronbach's alpha of 0.92 for Q10 and 0.90 for Q3; as to the subscore for the first 13 items, they amounted to 94.6% for Q10 and 92.6% for Q3; the subscore of the seven last items amounted to 87% for Q10 and 83.2% for Q3. This shows that patients remained consistent with their answers and that neither questionnaire was more reliable than the other. Furthermore, it can be observed that the first 13 items of both Q10 and Q3 have a better internal coherence than the last 7 items and the complete questionnaires.

Factor analysis (FA) was performed separately for Q10 and Q3 questionnaires (added in the statistical

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TABLE II. MSGS Salivary Score Q10 (0–10 Scale Version).		
• Measure	• Items	
Dry mouth	Since 1 mo	
	1. Evaluate the intensity of your mouth drynessNo dryness 0 1 2 3 4 5 6 7 8 9 10 Maximal dryness	
Salivary glands	2. Evaluate the frequency of your mouth dryness during the dayNever 0 1 2 3 4 5 6 7 8 9 10 Constantly during the day	
	3. Evaluate the quality of your salivarnormal (even if diminished) 0 1 2 3 4 5 6 7 8 9 10 very thick/sticky/watery (serous)/no saliva	
	4. Evaluate the taste of your salivaNormal 0 1 2 3 4 5 6 7 8 9 10 very salty and/or sweet and/or bitter and/or acid and/or bad taste	
	5. At which frequency do you feel the need to moisture your mouth during the day (either by drinking water / chewing gums / or by using moisturizing sprays)?Never 0 1 2 3 4 5 6 7 8 9 10 constantly	
	6. How frequently do you wake up at night to drink water? Never 0 1 2 3 4 5 6 7 8 9 10 Very frequently	
	7. Evaluate your talking difficulty related to your dry mouthNo difficulty 0 1 2 3 4 5 6 7 8 9 10 very important difficulty (constant need to moisturize to be able to speak)	
	8. Evaluate your level of difficulty to chew and swallow foodNo difficulty 0 1 2 3 4 5 6 7 8 9 10 very important difficulty (constant need to drink water to chew and swallow food)	
	9. Evaluate the dryness of your lipsNo dryness 0 1 2 3 4 5 6 7 8 9 10 maximal dryness	
	10. Evaluate the dryness of your noseNo dryness 0 1 2 3 4 5 6 7 8 9 10 maximal dryness	
	11. Evaluate the dryness of your eyesNo dryness 0 1 2 3 4 5 6 7 8 9 10 maximal dryness	
	12. Are you physical activities disturbed because of your dry mouth?No 0 1 2 3 4 5 6 7 8 9 10 yes, I avoid any activity which makes meUncomfortable because of my dry mouth	
	13. Evaluate your quality of life regarding to your dry mouthPerfect 0 1 2 3 4 5 6 7 8 9 10 completely unsatisfying	
	Since 1 mo	
	14. At what frequency do you experience a feeling of itching/tightness (tension) in the area of the salivary glands (in front of the ears and/or under the lower jaw)? Never 0 1 2 3 4 5 6 7 8 9 10 everyday	
	15. At what frequency do you experience swelling in the area of the salivary glands (in front of the ears and/or under the lower jaw) during meals?Never 0 1 2 3 4 5 6 7 8 9 10 at every meal	
	16. At what frequency do you experience swelling in the area of the salivary glands (in front of the ears and/or under the lower jaw) between meals?Never 0 1 2 3 4 5 6 7 8 9 10 at every meal	
	17. Evaluate the persistence of these swellingNo swelling 0 1 2 3 4 5 6 7 8 9 10 healing after a few months	
	18. How many times have you had to take antibiotics because of an infection of the salivary glands?Never 0 1 2 3 4 5 6 7 8 9 10 frequent infections (> one time per month)	
	19. Evaluate the pain caused by the salivary glands swellingNo pain 0 1 2 3 4 5 6 7 8 9 10 maximal pain	
	20. Evaluate the discomfort caused by the salivary glands swellingNo discomfort 0 1 2 3 4 5 6 7 8 9 10 maximal discomfort	

analysis section page 13). The clustering of the question corresponds well to our clinical grouping. Loading plot for the Q10 version revealed that the first two factors account for 89.2% of the variance. Loading plot for the Q3 version confirmed that the first two factors accounted for 88.6% of the variance. As questions 1, 2, 3, and 5 are those that contribute the most to the factor 1, we could assume that it concerns mostly xerostomia symptoms. Regarding factor 2, the last seven items are the ones that mostly contribute, meaning that it concerns mostly sialadenitis symptoms.

TABLE III. Participant's Characteristics by Diagnosis Compared With Healthy									
	Healthy N = 66	Stones N = 40	Idiopathic Sialositis N = 64	Sjögren's Syndrome N = 29	Other Sialositis (IgG4 Related Disease, Recurrent Parotitis) N = 6	Medication Related N = 9	Radiotherapy Treatment N = 42	Dysphagia, Nasal or Age Related N = 9	Total N = 265
Sex F	43 (65.2%)	23 (57.5%)	51 (79.7%)	27 (93.1%)	6 (100%)	7 (77.8%)	16 (38.1%)	6 (66.7%)	181 (68.3%)
Sex M	23 (34.8%)	17 (42.5%)	13 (20.3%)	2 (6.9%)	0 (0%)	2 (22.2%)	26 (61.9%)	3 (33.3%)	84 (31.7%)
Age (yr)	66.392 ± 13.3	40.478 ± 13.6*	57.503 ± 17.1**	19.493 ± 16.4*	5.461 ± 24.5	9.669 ± 15.6**	42.623 ± 12.0**	9.520 ± 14.8	265.505 ± 16.1
Total Q10 (range 0-200)	66.195 ± 29.3	34.367 ± 31.3	57.50 ± 30.7**	23.623 ± 37.1**	5.74 ± 35.6*	7.764 ± 33.3**	32.486 ± 33.3*	8.764 ± 34.1**	229.433 ± 36.6
Total Q3 (range 0-60)	60.65 ± 6.6	37.122 ± 9.4*	51.159 ± 8.7**	18.183 ± 10.2**	6.198 ± 12.6*	5.184 ± 10.5*	41.114 ± 7.1	4.283 ± 13.6**	224.123 ± 30.6
Q10 items 1 to 13 (range 0-130)	59.138 ± 6.6	32.182 ± 21.1	56.292 ± 26.4	18.556 ± 33.5**	5.472 ± 29.5	7.701 ± 20.3**	9.660 ± 35.1**	7.594 ± 31.6**	197.309 ± 30.6
Q3 items 1 to 13 (range 0-39)	58.43 ± 4.7	38.64 ± 6.5	49.87 ± 7.2	15.133 ± 7.4**	6.123 ± 9	3.173 ± 8.6**	6.132 ± 9.2	4.205 ± 11.2*	178.77 ± 7.7
Q10 items 14 to 20 (range 0-70)	58.339 ± 11.1	31.189 ± 19.9**	36.235 ± 15.1**	13.17 ± 17.8*	5.268 ± 21.6*	4.70 ± 14.0	3.00 ± 0.0	2.230 ± 12.7	161.135 ± 16.1
Q3 items 14 to 20 (range 0-21)	55.12 ± 3.3	38.68 ± 5.2**	42.75 ± 4.2**	16.57 ± 5.5*	6.78 ± 6.5	3.40 ± 6.1	3.07 ± 1.2	3.63 ± 3.1	166.43 ± 5.2
Salivometry lasting 15 min (ml/min)	16.033 ± 0.21	28.04 ± 0.15	50.42 ± 2.3	17.38 ± 2.4*	5.29 ± 1.5	3.007 ± 0.03*	2.015 ± 0.1	6.024 ± 0.03	166.43 ± 5.2
Salivometry lasting 6 min (ml/min)	10.059 ± 0.4	20.075 ± 0.3	26.34 ± 1.7	10.27 ± 1.2*	3.24 ± 1.5	3.012 ± 0.06*	No observations	2.03 ± 0.12	166.43 ± 5.2
Stenosis includes idiopathic stenosis (n = 65), Sjögren's syndrome (n = 28), IgG4 disease (n = 2), radioiodine therapy (n = 2), and juvenile recurrent parotitis (n = 2). N, mean ± SD.									
**P < .001.									

Regarding the time to complete the MSGS questionnaire, 99 patients and 26 healthy participants were recorded while completing the Q3, and they took respectively 5.23 min (±2.3 min) and 3.94 min (±3.94 min) versus 108 patients and 26 healthy volunteers who were recorded completing the Q10 and took, respectively, 5.65 min (±2.64 min) and 3.75 min (±2.11 min). Although not significant ($P = .193$), Q3 was completed slightly faster than Q10. The time to complete Q10 ($P = .0008$) and Q3 was significantly ($P = .0134$) shorter for healthy volunteers in comparison to patients.

Among the 38 participants who completed both questionnaires twice within 15 days, the coefficient of variation was 44.7% for Q10 and 41.5% for Q3, respectively, supporting the limitations of a subjective evaluation, as it is prone to vary among individuals.

The ICCs were not statistically significantly different between Q10 and Q3 regarding total scores and the sub-scores for questions 1–13 and 14–20, as their 95% CI overlaps. These ranged from 85% to 95% for total and questions 1–13 and were equal to 65% for the sub-score questions 14–20. We computed the individual difference between both repeats of the same questionnaire. To compare Q10 and Q3, we divided the difference by the maximum score obtainable by each questionnaire (Q3 = 60; Q10 = 200), expressed in percent. The average difference was 0.7% for

Q10 and 1.0% for Q3 and was not P statistically different ($= .3022$).

Regarding sialometry, 92 patients and 18 healthy volunteers underwent 6-min sialometry and had a mean salivary flow of, respectively, 3.32 g/min (±1.79 g/min) and 3.95 g/l (±2.56 g/min). The difference of sialometry at 6 min between patients and healthy volunteers was not statistically significant ($P = .2035$). One-hundred and fifty-seven patients and 24 healthy volunteers underwent a 15-min sialometry and had a mean salivary flow of, respectively, 4.2 g/min (±2.33 g/min) and 5.7 g/min (±3.33 g/min). The difference of the sialometry at 15' min between patients and healthy volunteers was statistically significant ($P = .0067$) and revealed higher values for healthy volunteers in comparison to patients.

Healthy participants had significantly lower Q10 and Q3 scores and higher sialometry values at 15 minutes than the pathological groups. The difference of sialometry values was significant only between healthy volunteers and patients at 15 minutes, possibly because the former need time to produce enough saliva to differentiate them from patients. Moreover, the number of healthy volunteers was limited, which implies that we need to consider those results with caution.

Healthy volunteers had a mean Q10 score of 19.6 (±23.4) and a mean Q3 score of 5.4 (±6.4) in contrast to

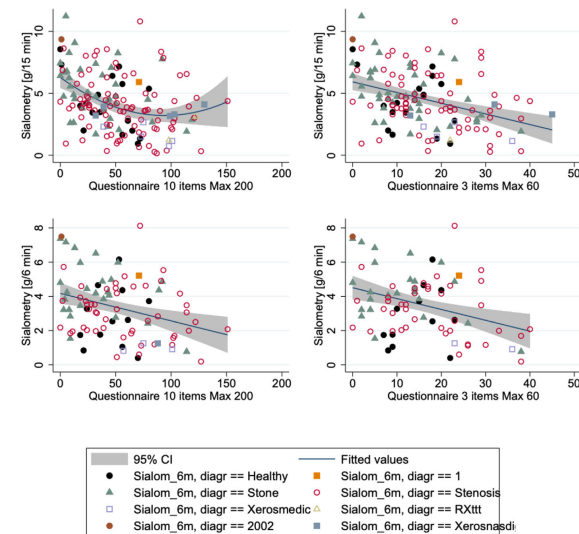


Fig. 1. Regression curve/line of the association between our questionnaires (Q3 and Q10) and sialometry. [Color figure can be viewed in the online issue, which is available at www.laryngoscope.com.]

patients who had a mean Q10 score of 52.6 (± 33.8) and a mean Q3 score of 14.9 (± 9.3). Patients had higher scores compared to healthy volunteers, and the difference was statistically significant for Q10 ($P = .049$) and Q3 ($P = .046$).

To further validate the questionnaire, patients with stenotic and calculous sialadenitis were analyzed using the first 13 and the last 7 questions separately. Regarding the group of patients with stenosis, assessment with the Q10 ($n = 307$) 1–13 items subscore revealed no association ($R^2 = 1.5\%$, $P = .110$). However, a better association was found with Q3 ($n = 290$), although not sufficient to predict a stenosis ($R^2 = 2.4\%$, $P = .0018$). Regarding the group of patients with calculous sialadenitis, assessment with the Q10 14–20 items subscore revealed a low positive association for calculus ($n = 254$, $R^2 = 1.5\%$, $P = .036$, OR = 1.019, 95% CI = 1.001–1.04), which was slightly better with Q3 ($n = 263$, $R^2 = 2.4\%$, $P = .007$, OR = 1.075, 95% CI = 1.02–1.12). An OR of 1.075 means that for a one-point increase in the Q3 questionnaire, the risk of stenosis increases by 7.5%. The positive association of Q3 has a higher correlation in the presence of calculus and stenosis compared to Q10. The scores did not predict the presence of stenosis or calculus, thus supporting the notion that the value of the questionnaire lies in quantifying the degree of salivary symptoms and not predicting a particular disease.

Regarding the correlation between the MSGS questionnaire scores and the sialometry results, as shown in Figure 1, the amount of saliva produced at 6 minutes was significantly associated with Q10 ($N = 87$; $P = .0038$; adjusted $R^2 = 8.4\%$) and with Q3 using a linear model ($P = .0013$, adjusted $R^2 = 11.5\%$). At 15 minutes, Q10 was also significantly associated with the linear model, explaining 10.3% of the variance ($N = 139$; $P = .0001$; adjusted $R^2 = 10.3\%$). At 15 minutes, Q3 with the linear model explained 13.6% ($P = .0001$; adjusted $R^2 = 13.6\%$). However, the coefficient of determinations R^2 being below 80% precludes any individual prediction of the sialometry results. From logistic regression, none of the questionnaires (total score or subscore 1–13) was associated with pathological sialometry at 6 and 15 minutes (OR not statistically different from 1.0; P -value ranging from .051 to .08), supporting the importance of an objective evaluation to assess hyposalivation.

DISCUSSION

Many subjective questionnaires assessing symptoms and QOL regarding xerostomia and salivary gland diseases exist in the literature. Most of them assess xerostomia, sialadenitis, or QOL. There is a need for a comprehensive and standardized questionnaire to characterize the symptoms of patients experiencing xerostomia and/or sialadenitis and to evaluate the effect of all treatments.

Regarding QOL, three questionnaires are frequently addressed in the literature: the Health Survey (SF-36, SF-8),^{21,22} which evaluates the QOL in regard to the patient's general health; the Glasgow Benefit Inventory,²³ which assesses QOL in regard of treatment in the

otolaryngology field; and the Oral Health Impact Profile Questionnaire,^{24,25} which evaluates QOL in regard to the patient's oral health. These scores evaluate the QOL of patients but lack specific features regarding the symptoms of xerostomia and sialadenitis. The purpose of the MSGS questionnaire is to target salivary gland pathology symptoms and their consequences on the general QOL. We believed that our questionnaire needed to be exhaustive regarding salivary gland symptoms but also as concise as possible to be able to be used in clinical practice. Therefore, we added only four specific items regarding QOL (questions 12, 13, 19, and 20). Also, we did not want to complicate the study by asking the patients to fill another questionnaire focused only on QOL, as our aim was to evaluate the overall salivary gland burden.

Regarding salivary gland symptoms, two questionnaires address both xerostomia and non-tumoral salivary gland pathologies, and both have been systematically validated: the Xerostomia Inventory (XI)²⁶ and the Chronic Obstructive Sialadenitis Score (COSS).²⁷

XI²⁶ was first published in 1999 and comprises 11 questions on xerostomia and 4 additional items regarding the burning-mouth syndrome. Although XI comprises many aspects related to dry mouth and has been validated and widely used in the literature,^{28,29} it is not exhaustive regarding xerostomia symptoms. Because it does not evaluate the ability to speak, potential taste disturbances, and saliva quality, it does not address QOL and does not evaluate at all sialadenitis features.

The COSS questionnaire²⁷ was proposed in 2016 to assess using 20 questions the effect of sialendoscopy on general salivary pathologies and dry mouth. It is the first questionnaire addressing sialadenitis features. Since its publication, it has been updated using the results obtained with pre- and post-operative assessment.^{30–32} COSS is able to precisely assess sialadenitis features, especially before and after sialendoscopy; however, it remains incomplete to evaluate other pathologies leading to xerostomia and treatment effects.

To validate the MSGS questionnaire, 199 patients and 66 healthy volunteers participated. The control group had significantly lower MSGS questionnaire results than the patient group, supporting its ability to distinguish between pathological and healthy participants. Sialometry at 6 and 15 minutes was also assessed on both groups. The results showed a significant negative association between Q3/Q10 questionnaires and sialometry results, but the amount of variance explained was not sufficient to allow individual predictions of sialometry values and therefore to detect hyposalivation.

We assessed two versions of the MSGS questionnaire (Q3 and Q10) to evaluate whether fixed answers would add more objectivity, allowing better reproducibility of the score over time; however, no differences were found between the answers of patients with Q3 and Q10 questionnaires when completed within an interval of several days. Moreover, the consistency of both versions, measured by ICC, were similar for both Q3 and Q10. Both versions of the MSGS questionnaire showed good internal coherence, confirming that the answers to the

questions were consistent among themselves. This was even more accentuated for the first 13 items of the questionnaires.

The time to complete both versions of the MSGS questionnaire was assessed, and no significant difference was found. Moreover, to our knowledge, the time required to fill out questionnaires regarding xerostomia and salivary gland pathologies has not been investigated in the literature.

As both versions Q3 and Q10 remain statistically similar regarding their scale reliability, their ability to distinguish pathological groups from healthy controls, and time to fill out, both would be adequate for use. However, in order to have only one questionnaire as a consensus, we recommend the use of the Q10 version. First, the 1–10 visual analogue scale is widely used and accepted in the literature and has the reputation to be better understood by patients. Second, the Q10 version will be more suitable for an international use, as its translation in different languages might be more accurate than of the Q3 version.

Our study has some limitations. First, we chose to use a less common approach to collect saliva, in contrast to the more frequent methods described by Navazesh et al., as we believed the swab method to be easier to carry out with patients with dry mouth. Based on our clinical experience, we believed it to be very difficult to ask patient with xerostomia to spit or drain saliva from a dry mouth. Moreover, our study had a restricted number of participants, especially in regard of radioiodine therapy and auto-immune salivary gland pathologies such as Sjögren's syndrome, IgG4 disease, and radioiodine.

Therefore, future multicentric studies are needed. The translation of the MSGS questionnaire in other languages and the initiation of a prospective multicentric study using the MSGS questionnaire have been decided during the second International Sialendoscopy Society Meeting. This attempt to use in a multicentric setting a standardized questionnaire in centers dealing with salivary gland diseases may facilitate its improvement and allow a greater consensus in the future. As for an immediate result, it will allow carrying out comparative clinical studies on various treatment modalities (e.g., medical, sialendoscopic, or surgical interventions), comparing pre- and post-treatment scoring. Hopefully, in the future, a wide consensus around a unique screening and assessment tool might be meaningful for all patients suffering from salivary gland diseases.

CONCLUSION

We proposed here a novel, reliable, comprehensive, and self-administered questionnaire addressing benign salivary gland lesions such as xerostomia, sialadenitis, and the associated QOL of the patients. These questionnaires have been statistically validated on a series of patients affected by different pathologies. Even though our questionnaire was not able to predict the presence of stones or strictures, it could discriminate patients from

healthy volunteers and demonstrate a good internal coherence of its items. Both versions Q3 and Q10 of the MSGS questionnaire were analyzed and proved to be similar regarding their scale reliability, their ability to distinguish pathological groups from healthy controls, and time to fill out.

We recommend the use of the Q10 version of the MSGS questionnaire, as its scale is better understood by patients and clinicians. In addition, it is more suitable for translation, as we aim to implement its use for international studies.

Further multicentric investigations are in progress to validate and improve it with a larger cohort of patients in 27 different countries.

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DISCUSSION

Notre étude publiée dans le journal « Laryngoscope » est la première validant un questionnaire de consensus afin d'évaluer les pathologies salivaires non tumorales et la sécheresse orale. Il paraissait important de créer un questionnaire qui puisse inclure une évaluation à la fois de la sécheresse orale, mais aussi des symptômes liés à la sialadénite. Les patients peuvent se présenter avec un mélange de symptômes salivaires qu'il est important de pouvoir quantifier afin de pouvoir établir un diagnostic et une prise en charge adéquate. De plus la sialadénite est une affection qui comporte des options thérapeutiques médicamenteuses et chirurgicales. L'avènement de la sialendoscopie a prouvé son efficacité dans le traitement de la sialadénite, que ce soit par la dilatation de sténoses présentes au niveau des canaux salivaires (1) et / ou grâce à l'extraction de calculs salivaires (2). Cependant la quantification précise de la symptomatologie avant et après traitement reste encore peu répandue.

Le questionnaire MSGS a été validé statistiquement grâce à 199 patients et 66 volontaires, et accepté par la majorité des membres de la Société Internationale Multidisciplinaire des glandes salivaires comme questionnaire de référence.

Ce questionnaire de 20 questions comporte deux parties : la première en relation avec des symptômes de sécheresse orale et la deuxième en rapport avec des manifestations de sialadénite. Avant de remplir le questionnaire, des questions anamnestiques ciblées doivent être posées au patient, comprenant la consommation tabagique, la prise de traitements avec effet sur la sécrétion salivaire, un ou plusieurs diagnostics connus de maladies systémiques spécifiques (syndrome de Sjögren, maladie liée aux IgG4, parodontite juvénile récurrente par exemple) et un antécédent de traitement par iode radioactif ou radiothérapie cervico-faciale.

Les treize premières questions concernant la xérostomie portent sur l'intensité et la fréquence de la sécheresse orale, la qualité et le goût de la salive, le besoin de s'hydrater pendant la journée et la nuit, ainsi que des difficultés à l'élocution et l'alimentation ; ces deux derniers éléments étant les conséquences les plus invalidantes de la sécheresse orale dans la vie quotidienne. La sécheresse des lèvres, du nez et des yeux est aussi évaluée dans ce nouveau questionnaire afin de pouvoir orienter l'hypothèse étiologique vers certaines maladies immunologiques ou effets secondaires de médicaments ou traitement par radio-iode. La qualité de vie des patients en rapport avec la symptomatologie est aussi explorée par ce questionnaire.

Les sept dernières questions concernent l'évaluation d'une éventuelle sialadénite à travers des symptômes de tension ou de tuméfaction des glandes salivaires pendant et/ou en dehors des repas. Il est à noter que la présence de tensions ou tuméfactions des glandes salivaires pendant les repas est typique d'une sialadénite aigüe, tandis que ces mêmes symptômes, présents aussi ou uniquement en

dehors des repas, témoignent d'une sialadénite chronique. La persistance de ces manifestations dans le temps accompagnées ou non du besoin de prendre des antibiotiques sont aussi recherchés, et permettent ainsi une comparaison avant et après traitement.

Les dernières questions concernant la douleur et l'inconfort provoqués par ces tensions et/ou tuméfactions des glandes salivaires ont pour but d'estimer la qualité de vie de ces patients.

Jusqu'à présent, le COSS (3) était l'unique questionnaire évaluant les symptômes de sécheresse orale et de sialadénite. L'objectif du COSS était de quantifier une éventuelle amélioration ou péjoration des symptômes de sialadénite notamment après un traitement par sialendoscopie (4). Il nous a paru dès lors intéressant de le comparer avec notre questionnaire MSGS.

En lien avec les symptômes de sécheresse orale, le COSS évalue uniquement la fréquence de la xérostomie et non son intensité comme c'est le cas de notre questionnaire MSGS. L'intensité de la xérostomie est essentielle pour estimer sa sévérité et donc ses conséquences sur la qualité de vie du patient, mais aussi afin de la corrélérer avec des mesures objectives d'hyposalivation comme la sialométrie. De plus, le questionnaire MSGS analyse la qualité ou consistance de la salive, non mentionnée dans le COSS, permettant de refléter la gravité de la sécheresse buccale en cas de xérostomie médicamenteuse, de sialadénite chronique ou de radiothérapie. Le COSS n'évalue pas le besoin d'hydratation de la bouche, or c'est une conséquence directe de la sécheresse orale. La capacité à mastiquer et avaler est évaluée en deux questions différentes dans le COSS. Nous avons décidé de combiner ces deux éléments en une seule question dans notre questionnaire MSGS car ils nous paraissent étroitement liés. Ainsi des difficultés à mâcher avec une déglutition normale sont plutôt en rapport avec des pathologies dentaires ou buccales, tandis que des difficultés de déglutition en l'absence de plaintes de mastication sont plutôt à mettre en rapport à d'autres troubles oto-rhino-laryngologiques sans lien avec une sécheresse orale. Le COSS comporte une question concernant un éventuel excès de salive. Nous n'avons pas jugé important de l'ajouter à notre questionnaire MSGS étant donné que les hypersalivations sont peu fréquentes et qu'elles n'étaient pas le but de notre étude. Le COSS ne comporte pas de questions concernant la sécheresse des lèvres, du nez et de yeux, essentielles pourtant pour mettre en évidence une éventuelle pathologie systémique immunologique, les effets secondaires de médicaments ou les conséquences d'un traitement par radio-iode.

En lien avec les symptômes de sialadénite, le COSS ne mentionne pas les symptômes de tension et/ou démangeaisons dans la région des glandes salivaires, or ce sont des manifestations fréquentes de sialadénite chronique due aux sténoses. Les tuméfactions récurrentes des glandes salivaires pendant et entre les repas étant fréquemment associées à une sialadénite chronique, il nous paraissait plus utile d'effectuer cette distinction que d'aborder l'élément de « gène » lorsque l'on appuie ou non sur la région des glandes salivaires comme effectué dans le COSS. A la différence du COSS qui n'analyse pas

une éventuelle persistance des gonflements ainsi que la nécessité de prendre ou non des antibiotiques, nous avons considéré ces éléments comme étant des indicateurs de gravité étant donné que la sialadénite inflammatoire peut se compliquer par une infection.

Notre questionnaire MSGS avait été créé sous deux formats : le premier comportant des réponses de 0 à 10 sous la forme d'une échelle visuelle analogique et le deuxième ayant quatre réponses détaillées pour chaque question. Pour le deuxième format, l'hypothèse d'un questionnaire ayant des réponses proposées était de limiter les biais en guidant les patients vers les réponses qui correspondaient le plus à leurs symptômes et d'améliorer la compréhension des questions. Toutefois l'analyse statistique comparant les deux versions des questionnaires n'a pas révélé de différence significative quant aux réponses données par les patients, ainsi qu'au temps nécessaire pour les remplir.

L'utilisation de ce questionnaire se fera dans plusieurs pays et sera ainsi traduit en plusieurs langues étrangères. Afin de limiter les biais de traduction, et tout en sachant que le format d'échelle visuelle analogique est largement utilisé et accepté dans la littérature, il a été décidé de manière consensuelle de l'implémenter sous cette version. De nouvelles études multicentriques prospectives sont en cours dans divers pays afin de le valider avec un plus grand nombre de patients et selon les résultats il est possible que des modifications soient apportées. Dans l'immédiat, l'utilisation de notre questionnaire MSGS permettra de réaliser des études cliniques comparatives quant aux diverses modalités de traitements des pathologies salivaires et de la xérostomie, en comparant le score avant et après traitement.

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